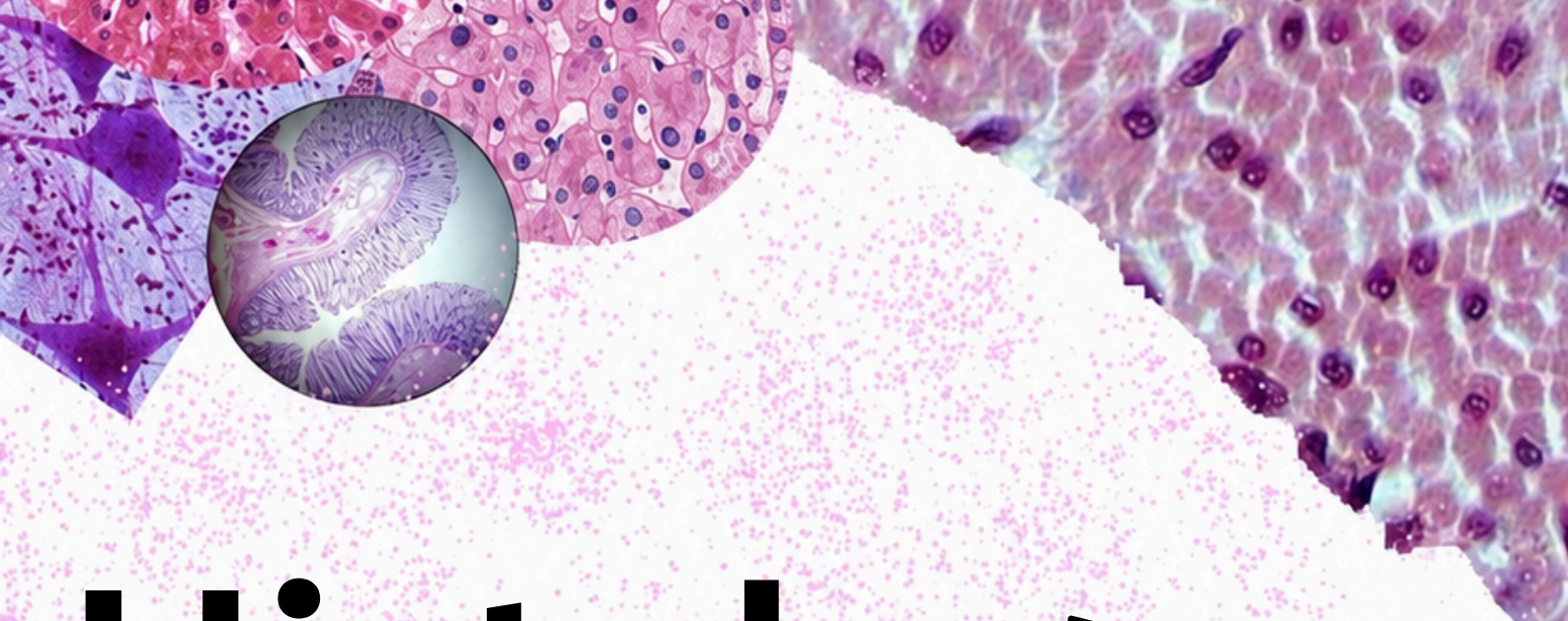




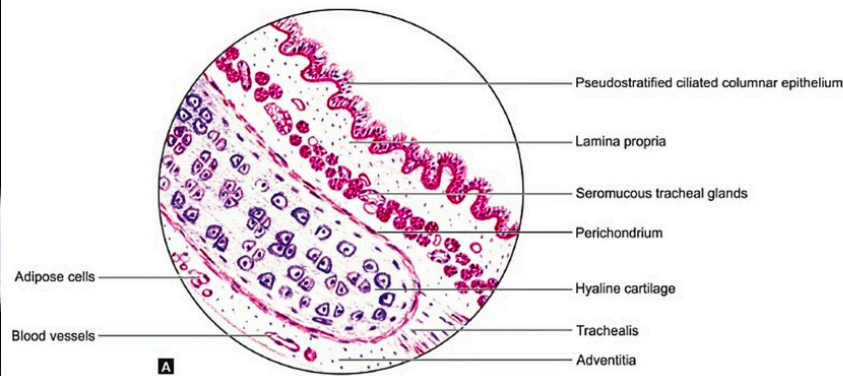
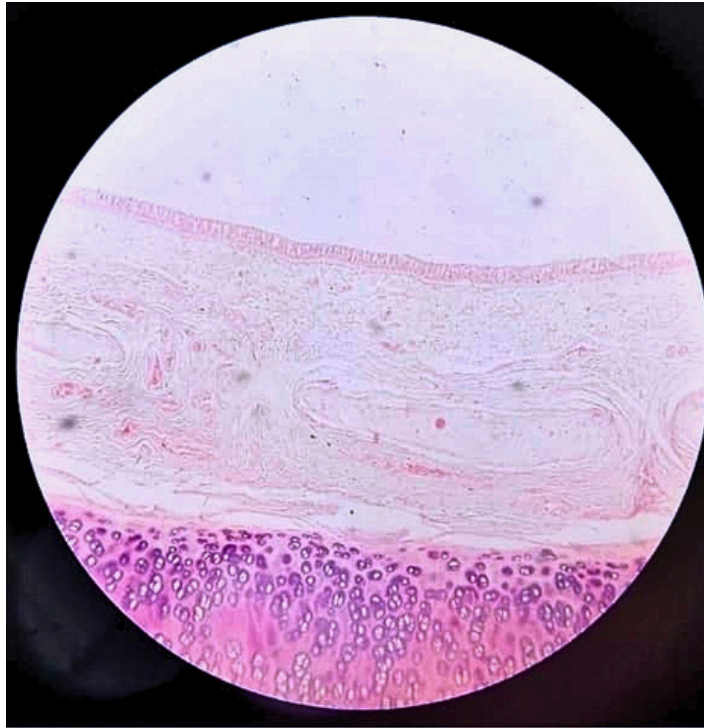
# Histology Slides

-by Mission Distinction



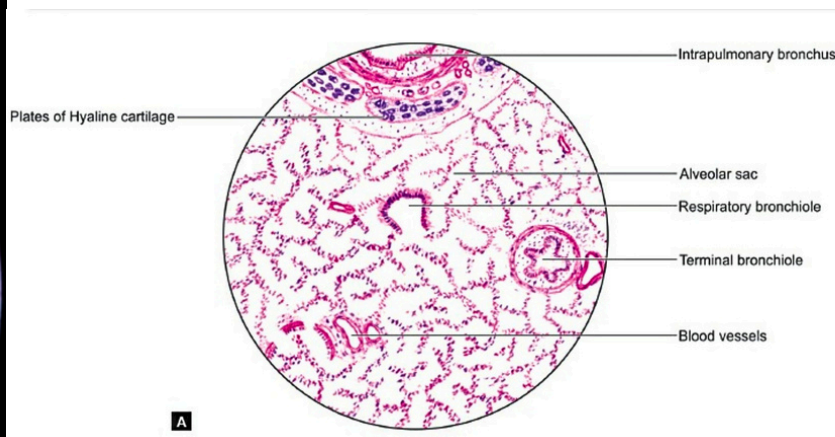
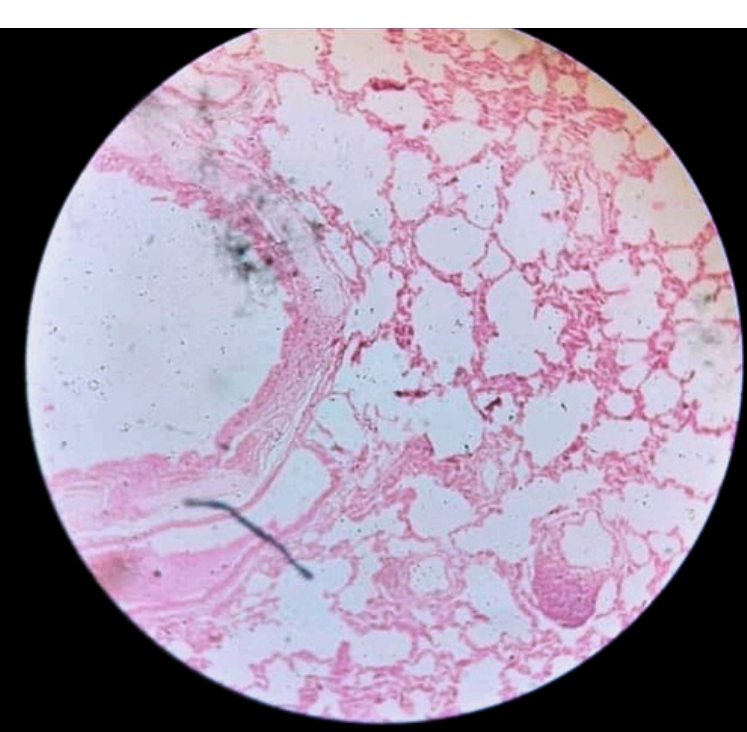


**\* Note - Red highlighted points are identifying features for Spotting, others are for commenting slide**



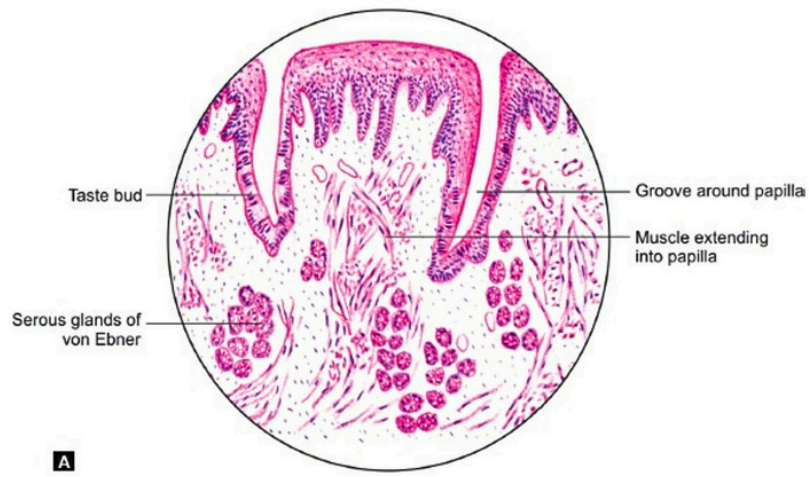
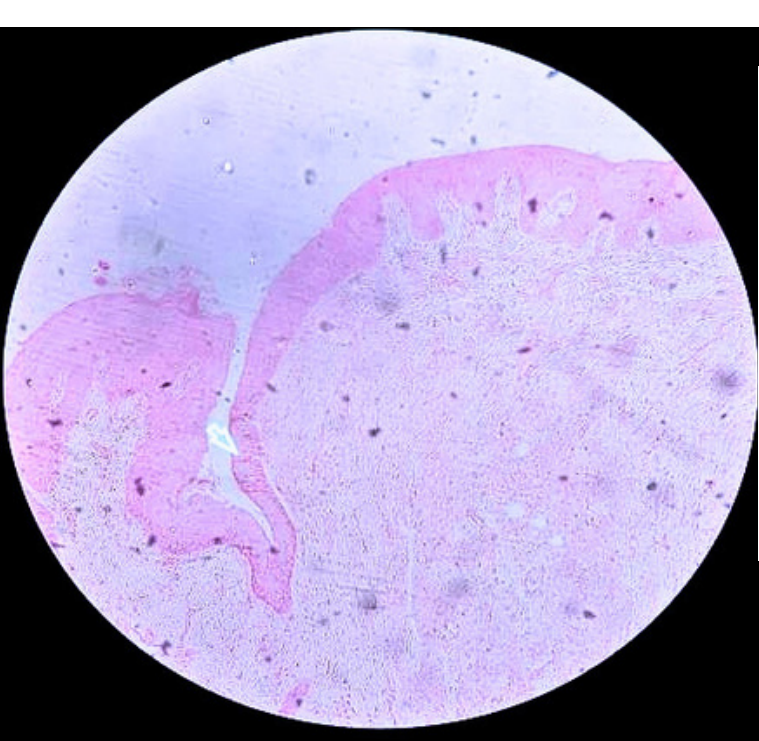
# Trachea

- **Lining Epithelium: Pseudostratified ciliated columnar epithelium**, predominantly composed of ciliated cells and mucus-secreting goblet/mucous cells.
- Basement Membrane: Notably thick, glassy, and homogeneous in appearance.
- Submucosal Glands: The submucosa features seromucous glands that keep the epithelium moist and trap dust particles.
- Cartilage Type: The skeletal framework consists of 16 to 20 C-shaped rings of **hyaline cartilage**.
- **Trachealis Muscle**: The posterior open end of the C-shaped cartilage is bridged by smooth muscle called the trachealis muscle.
- Adventitia: The outermost layer is made of fibroelastic connective tissue housing blood vessels and nerves.



# Lungs

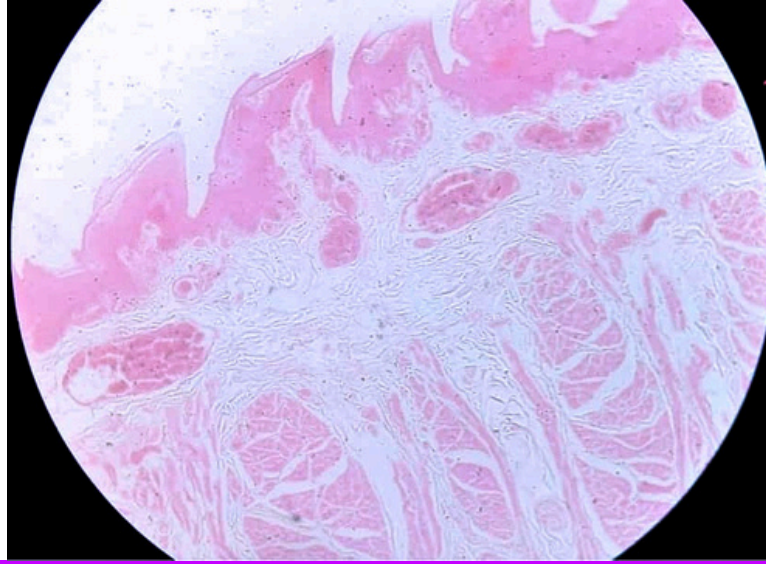
- **Alveolar sac lined by Simple Squamous Epithelium**
- **Respiratory bronchioles lined by simple cuboidal epithelium**
- Bronchus & **Bronchiole** difference - Bronchus has irregular plates of hyaline cartilage and submucosal glands, whereas both are completely absent in a bronchiole.
- Type I **Pneumocytes**: Simple squamous cells that line 90% of the alveolar surface and form an integral part of the blood-air barrier for gas exchange.
- Type II Pneumocytes: Rounded, secretory cuboidal cells with lamellar bodies that secrete pulmonary surfactant to reduce surface tension.
- Dust Cells: Alveolar macrophages present in the connective tissue or alveolar lumen that phagocytose inhaled dust particles and debris.



## Tongue (Circumvallate Papillae)

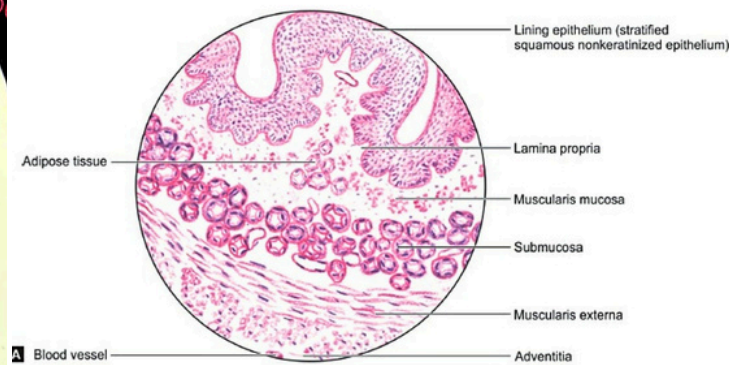
- **Epithelium:** Covered by stratified squamous non-keratinized epithelium .
- **Taste Buds:** Numerous pale-staining, barrel-shaped taste buds are clearly visible in the lateral walls of the papilla facing the trench.
- **Von Ebner's Glands :** Purely serous salivary glands are present deep in the lamina propria, opening into the bottom of the trench to wash away food particles.
- **Core:** A central core of connective tissue (lamina propria) forms secondary papillae that indent the epithelium.
- **Muscular Base:** The deep layer consists of interlacing bundles of skeletal muscle running in three different planes (longitudinal, transverse, and vertical).





## Tongue (Filiform & Fungiform Papillae)

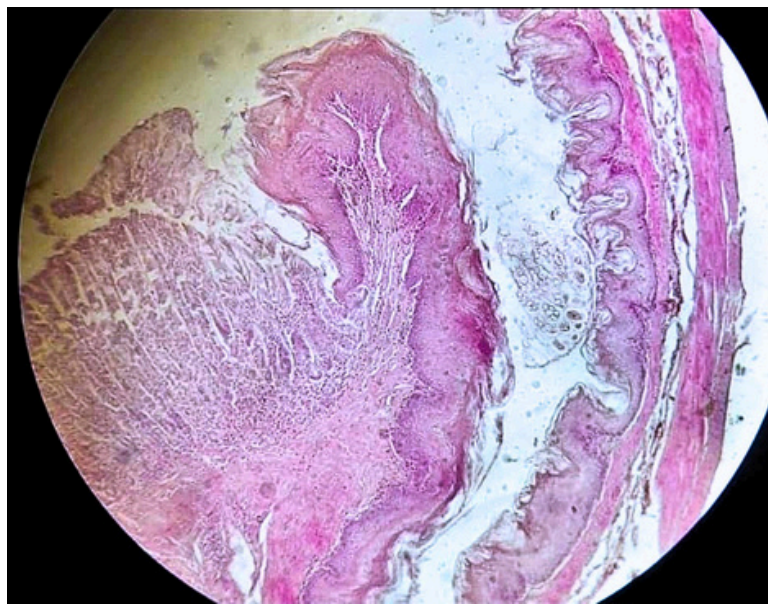
- **Filiform Papillae** Structure: Presents numerous small, conical, or pointed projections lined by stratified squamous epithelium.
- Filiform Keratinization: The epithelium covering the tips of the filiform papillae is heavily keratinized to provide a rough surface for mechanical scraping.
- **Fungiform Epithelium & Taste Buds**: Covered by **non-keratinized stratified squamous epithelium**, occasionally showing a few pale-staining taste buds on their upper apical surface.
- Striated Muscle Core: Beneath the lamina propria lies a dense mass of interlacing skeletal muscle fibers oriented in three perpendicular planes (vertical, transverse, and longitudinal).



# Oesophagus

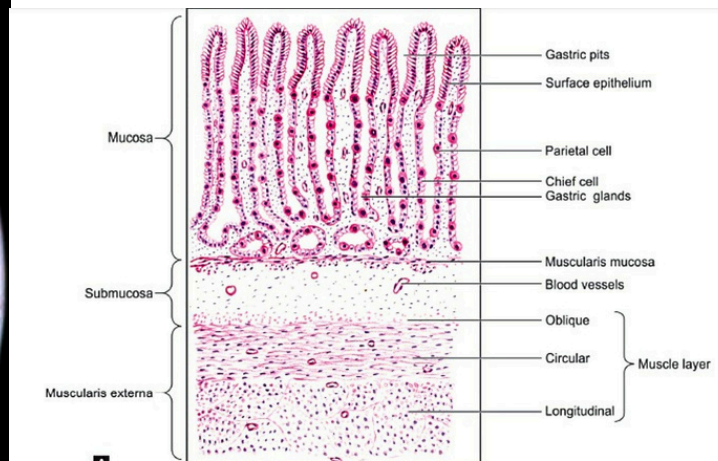
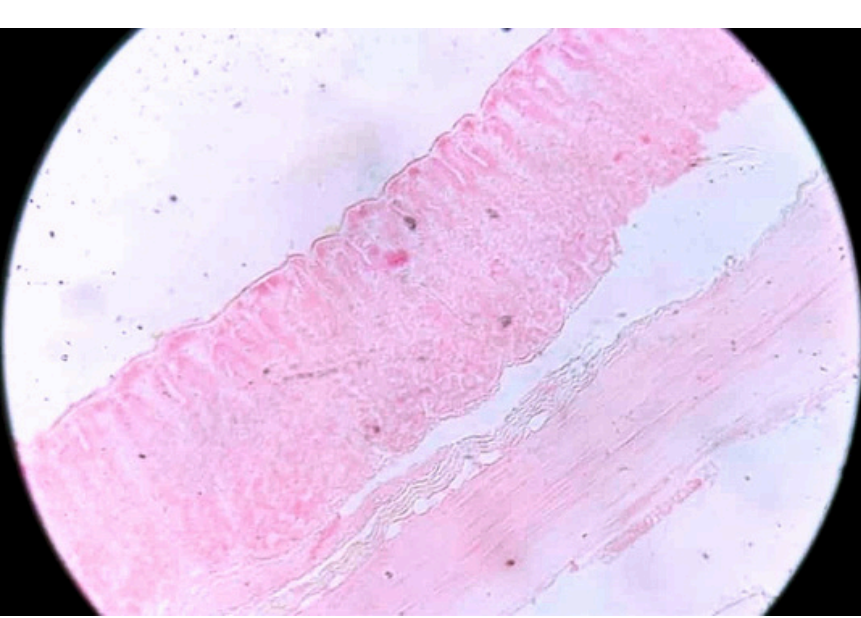
- **Mucosa Epithelium Lined by stratified squamous non-keratinized epithelium.**
- **Muscularis Mucosae:** Features a remarkably thick layer of longitudinally arranged smooth muscle fibers, which becomes progressively thicker in the lower parts.
- **Submucosal Glands:** The submucosa contains characteristic esophageal glands proper (mucus-secreting glands) to lubricate the lumen.
- **Muscularis Externa Composition:** Composed of two layers (inner circular, outer longitudinal) showing a mix of skeletal and smooth muscle depending on the level (skeletal in upper 1/3, mixed in middle 1/3, smooth in lower 1/3).
- **No Meissner's/Auerbach's Absence:** Standard enteric plexuses (Meissner's in submucosa and Auerbach's between muscle layers) are present to coordinate peristalsis.





## C-O Junction

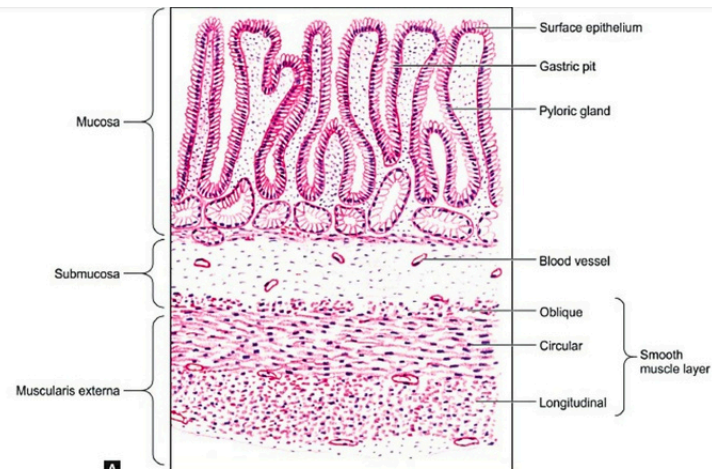
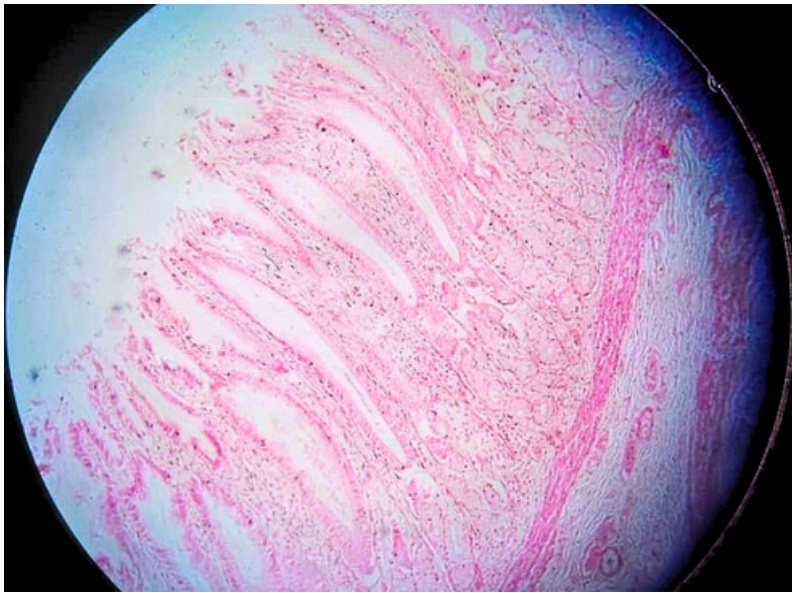
- Shows a sudden, distinctive **shift from** the **stratified squamous non-keratinized epithelium** of the esophagus to the **simple columnar epithelium** of the stomach (cardia).
- Lamina Propria Glands: The lamina propria changes rapidly to accommodate simple branched tubular cardiac glands that secrete protective mucus.
- Gastric Pits Appearance: The smooth surface of the esophageal zone abruptly gives way to the invaginations known as gastric pits on the stomach side.
- **Submucosal Glands Disappearance**: The characteristic esophageal glands proper present in the esophageal submucosa vanish as the section crosses over into the stomach territory.
- Muscularis Externa Thickening: The inner circular layer of the muscularis externa noticeably thickens at this zone to help form the physiological lower esophageal sphincter.



# Fundus of Stomach

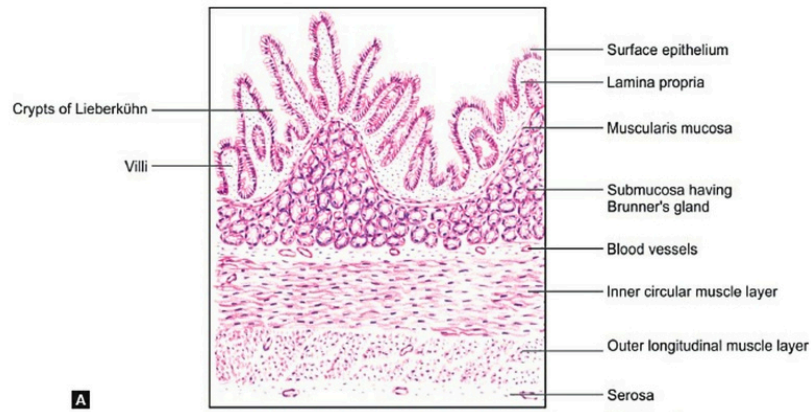
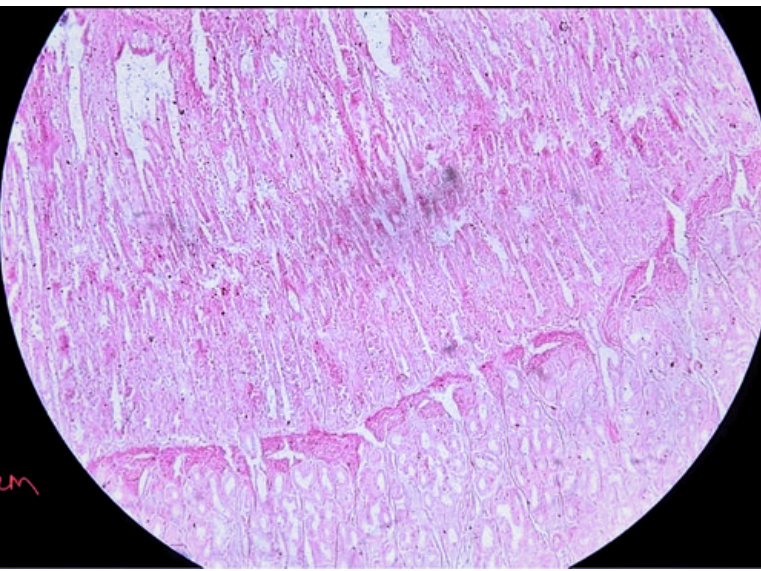
- **Lined by simple columnar mucus-secreting epithelium**
- **Gastric Glands:** The lamina propria is packed with long, straight, parallel gastric glands that open into the bottom of the gastric pits.
- **Chief (Peptic) Cells:** Deep in the bases of the glands are numerous chief cells, which are smaller, cuboidal/columnar, and strongly basophilic (bluish-purple); they secrete pepsinogen.
- **Muscularis Mucosae:** A thin but distinct layer of smooth muscle (muscularis mucosae) separates the mucosa from the submucosa.
- **Muscularis Externa Layers:** Features a thick muscular wall arranged in three layers of smooth muscle: an inner oblique, middle circular, and outer longitudinal layer.





# Pylorus of Stomach

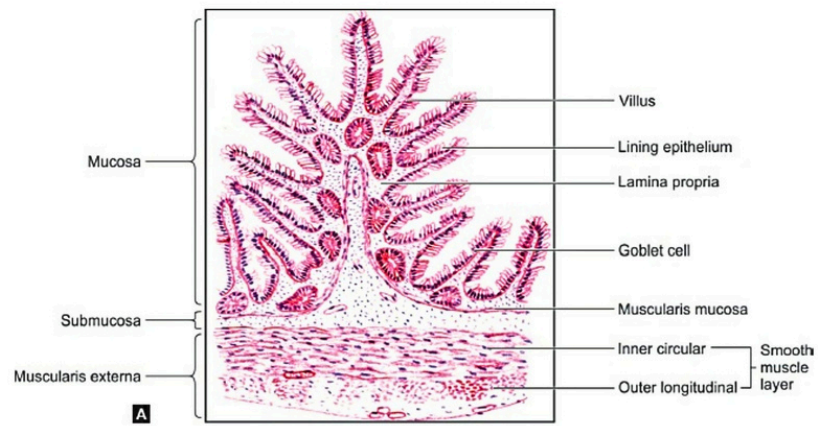
- **Lined by a tall simple columnar epithelium**
- **Deep Gastric Pits:** Unlike the fundus, the gastric pits are very deep and wide, occupying nearly half of the total mucosal thickness.
- **Branched Coiled Glands:** The lamina propria contains **pyloric glands** that are short, highly branched, and convoluted (coiled), rather than long and straight.
- **Thick Muscularis Externa:** The middle circular layer of the muscularis externa is remarkably thickened to form the anatomical pyloric sphincter.
- **No Villi or Submucosal Glands:** It can be differentiated from the duodenum by the complete absence of villi and the absence of Brunner's glands in the submucosa.



# Duodenum

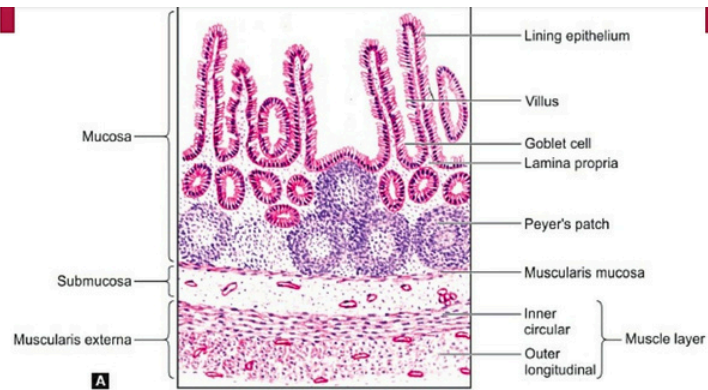
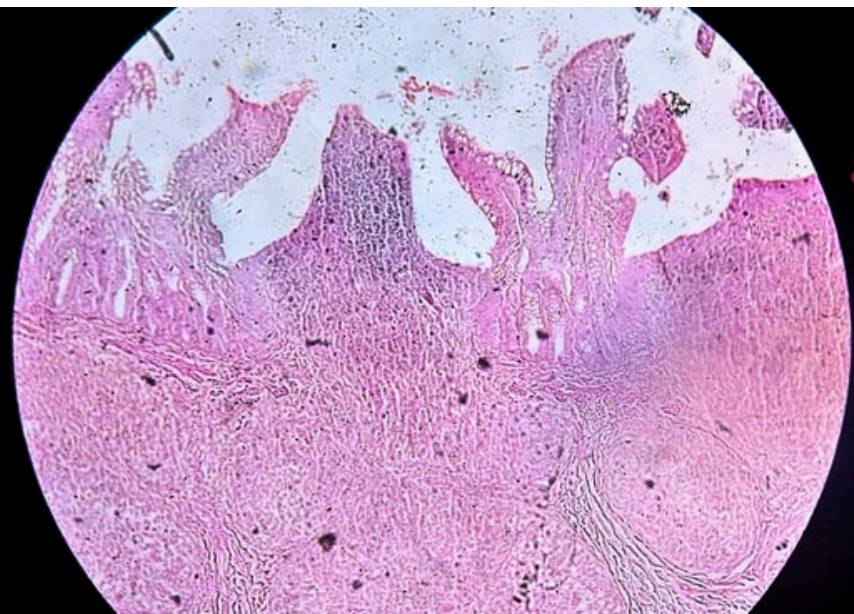
- **Lined by simple columnar epithelium**
- Presence of Villi: The mucosa is thrown into finger-like or leaf-like mucosal projections (villi) lined by simple columnar epithelium with a brush border.
- **Brunner's Glands** : The submucosa is completely packed with characteristic, pale-staining Brunner's glands (compound branched tubular mucous glands) that pierce the muscularis mucosae.
- Goblet Cells: The surface epithelium contains scattered, pale goblet cells
- **Paneth Cells**: The bases of the crypts contain distinctive Paneth cells with coarse, intensely acidophilic (bright red/pink) secretory granules.
- Muscularis Externa: Composed of standard inner circular and outer longitudinal layers of smooth muscle with Auerbach's plexus situated between them.





# Jejunum

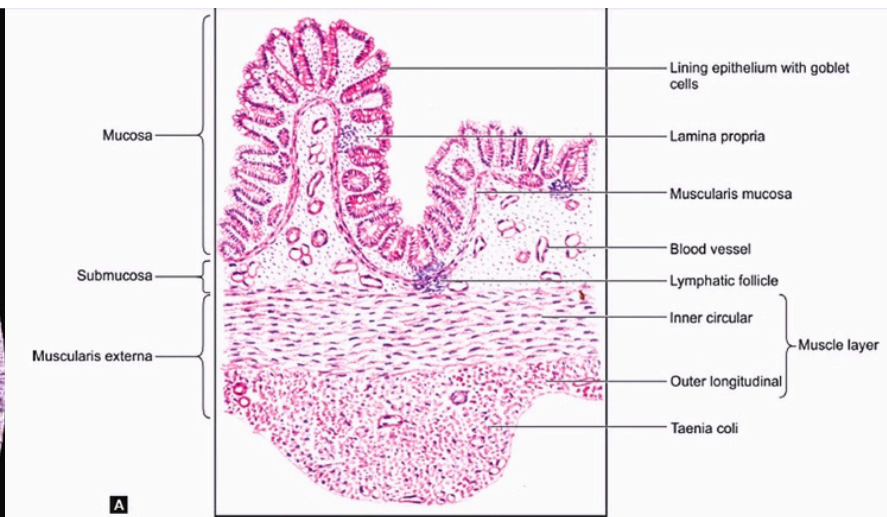
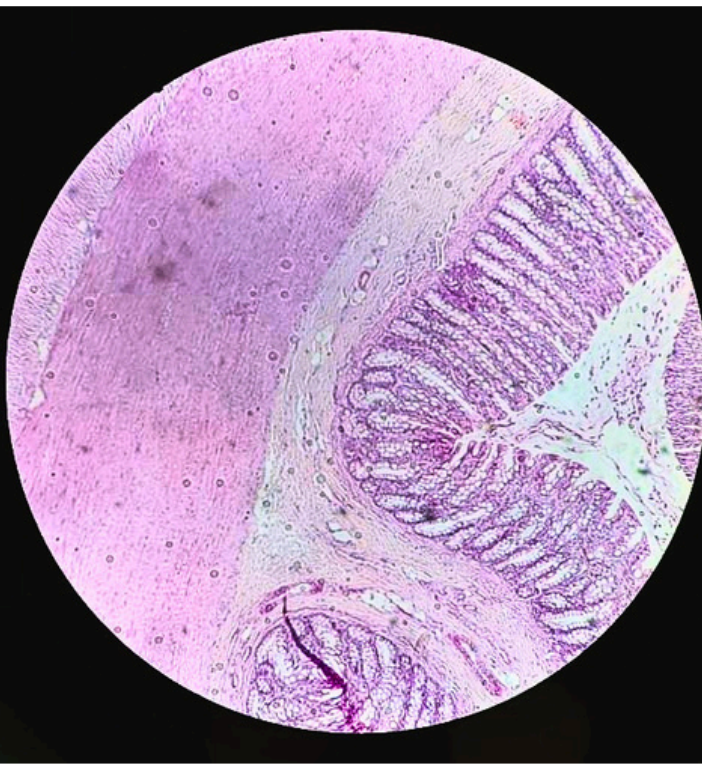
- **Mucosa lined by simple columnar epithelium** with goblet cells
- Tall, **Well-Developed Villi**: Features the tallest, largest, and most numerous finger-like villi among all parts of the small intestine to maximize the surface area for nutrient absorption.
- Plicae Circulares (Valves of Kerckring): The mucosa and submucosa are thrown into permanent, massive circular folds called plicae circulares, which are highly prominent in this section.
- Muscularis Externa & Serosa: The outer wall consists of two well-defined smooth muscle layers (inner circular, outer longitudinal) covered externally by a true serosa (visceral peritoneum).



# Ileum

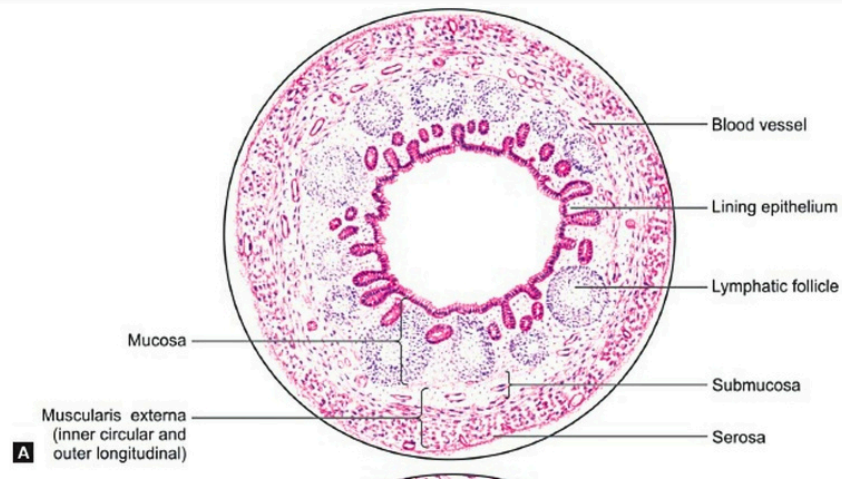
- **Lined by simple columnar epithelium**
- **Peyer's Patches** : The submucosa contains large, aggregated lymphoid follicles known as Peyer's patches that often breach the muscularis mucosae to extend into the lamina propria.
- **Abundant Goblet Cells**: The simple columnar surface epithelium contains a very high concentration of pale, mucus-secreting goblet cells (significantly more than in the duodenum or jejunum).
- **Crypts of Lieberkühn**: Intestinal glands (crypts of Lieberkühn) are present in the lamina propria between the bases of the short villi.
- **Muscularis Externa and Serosa**: Composed of standard inner circular and outer longitudinal smooth muscle layers, covered externally by a distinct visceral peritoneum (serosa).





## Large intestine

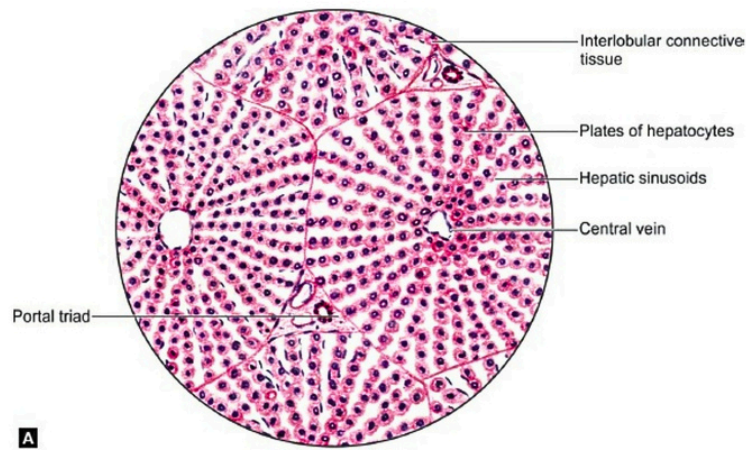
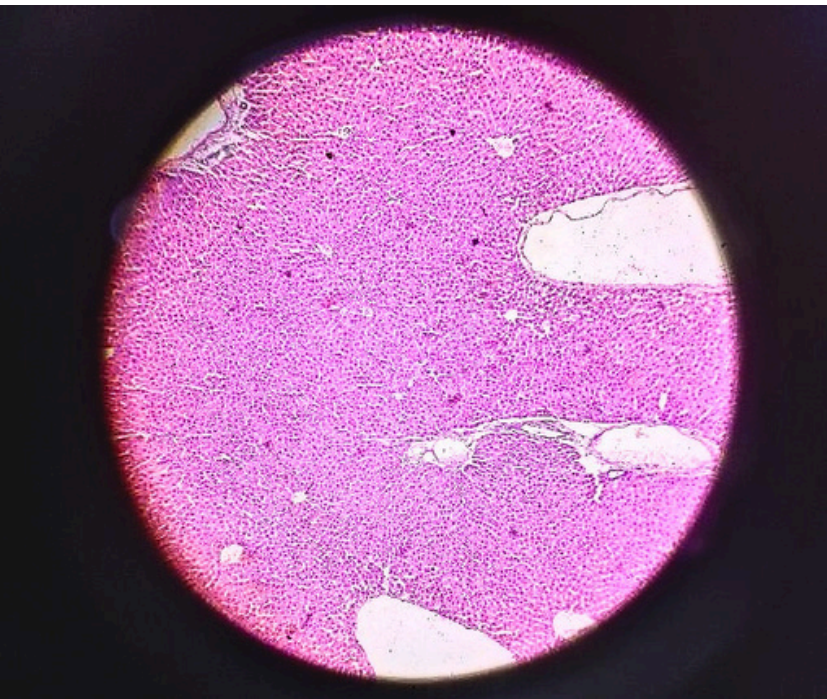
- **Lined by simple columnar epithelium**
- **Absence of Villi:** The mucous membrane displays crescent-shaped folds, but completely lacks villi.
- **Crypts of Lieberkühn:** The mucosa shows numerous closely arranged tubular glands (crypts), but notably lacks Paneth cells.
- **Abundant Goblet Cells:** The lining epithelium is rich in goblet cells, which increase in number caudally to provide lubrication for solidifying contents.
- **Muciphages in Submucosa:** The submucosa contains blood vessels, Meissner's plexus, and special cells containing PAS-positive granules called muciphages.
- **Taenia Coli:** The outer longitudinal muscle layer of the muscularis externa is aggregated into three thick bands called Taenia coli.



# Vermiform Appendix

- **Lined by simple columnar epithelium**
- **Lumen and Mucosa:** It features a small, narrow lumen with a mucosa devoid of villi and contains only poorly formed crypts.
- The submucosa is packed with abundant **lymphatic follicles** that extend into the lamina propria, earning it this nickname.
- **Muscularis Externa (No Taenia):** Consists of standard inner circular and outer longitudinal smooth muscle layers, but Taenia coli are completely absent.
- **Serosal Coverage:** The serosa covers the entire organ except at the attachment site of the mesoappendix.
- **Susceptibility to Infection:** Highly prone to appendicitis because it is a blind pouch prone to fecolith impaction, and it is supplied by an end artery making it vulnerable to gangrene.





# Liver

Each Hexagonal lobule has small round space in the centre known as **central vein**

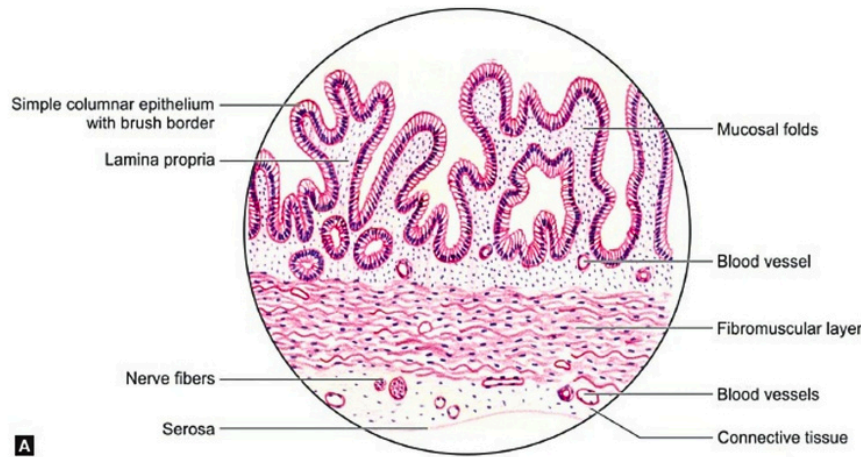
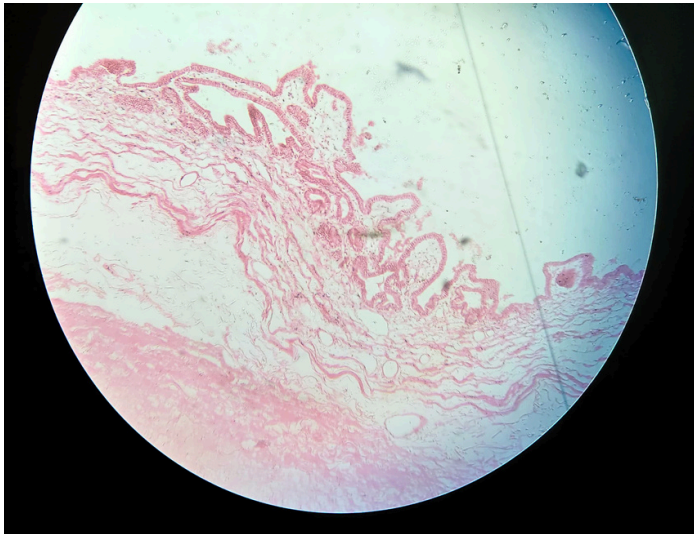
A number of broad irregular cords of cell seen to pass from these veins to the periphery of the lobule. These cords are made up of polygonal cells known as **hepatocytes**.

At regular interval between hepatocytes filled by connective tissue called Portal Canal

**Portal Triad** Composition: Located at the corners of the classical lobule, it contains a radicle of the portal vein, hepatic arteriole, and interlobular bile duct.

**Fenestrated Sinusoids & Kupffer Cells:** The spaces between hepatocyte plates contain fenestrated capillaries called sinusoids, which house phagocytic Kupffer cells.

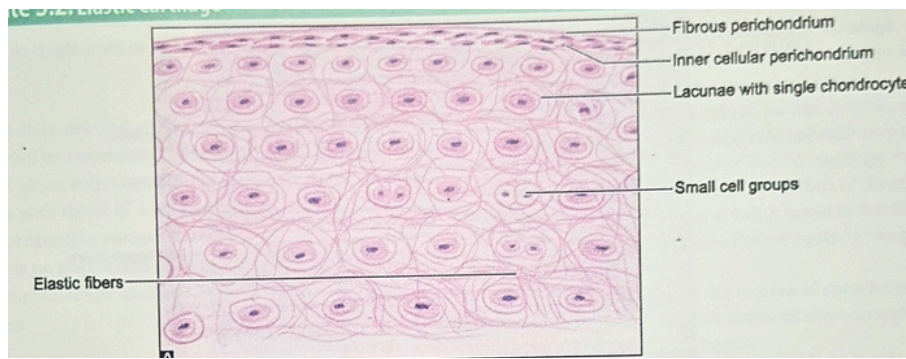
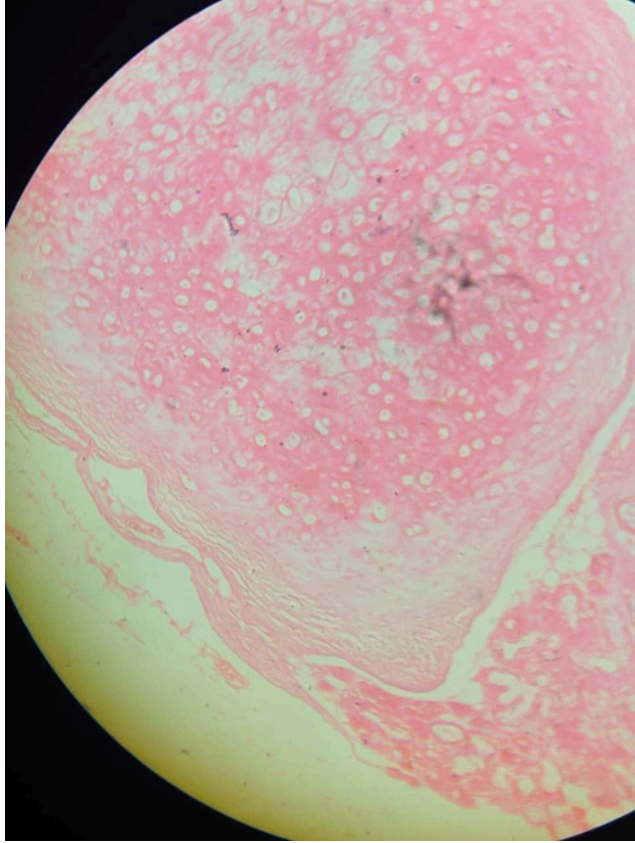
**Space of Disse:** A narrow perisinusoidal space between the endothelial lining of the sinusoid and the microvilli-lined surface of the hepatocytes.



# Gall Bladder

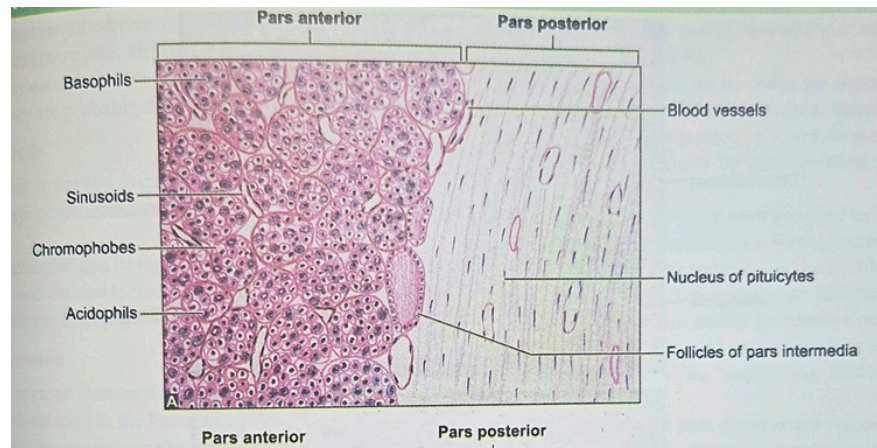
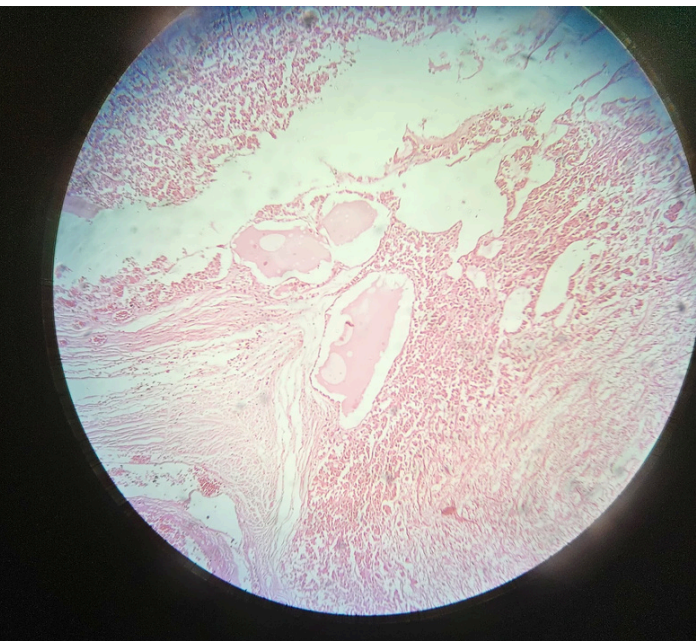
- The **mucous membrane is thrown into numerous branching** folds, giving the surface a characteristic honeycomb appearance.
- **Lining Epithelium:** It is lined by **simple tall columnar epithelium** with microvilli (striated border) and completely lacks goblet cells.
- **Deficient Mucosal Layers:** The mucosa contains only an epithelium and lamina propria; it uniquely lacks a muscularis mucosae and mucosal glands.
- **Fibromuscular Coat:** The muscularis externa consists of connective tissue mixed with smooth muscle fibers arranged in a random direction.
- **Absence of Lymphatics in Mucosa:** Unlike many parts of the gut, the lamina propria of the gallbladder is completely devoid of lymphatic vessels.





## Elastic Cartilage

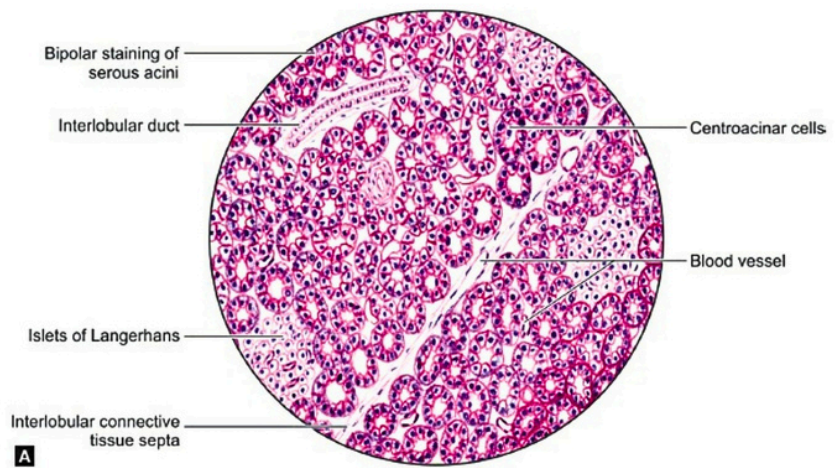
- **Presence of Perichondrium:** The outer boundaries are covered by a distinct layer of dense vascular connective tissue called the perichondrium.
- **Abundant Elastic Fibers :** The matrix is densely packed with a visible, branching network of elastic fibers that cluster heavily around the lacunae.
- **Chondrocytes in Lacunae:** Contains mature cartilage cells (chondrocytes) housed individually or in small groups inside spaces called lacunae.
- **Less Abundant Matrix:** Due to the dense concentration of cells and fibers, the amorphous ground substance/matrix is relatively sparse.
- matrix does not easily calcify with aging because it lacks the same structural layout.



# Pituitary Gland

- **Pars Distalis** Cellularity: The major part of the anterior pituitary (pars distalis) contains epithelial cells arranged in cords or clumps
- **Chromophils** strongly red-staining **acidophils** (somatotrophs, lactotrophs) and darkly bluish-purple-staining **basophils** (gonadotrophs, thyrotrophs, corticotrophs).
- Chromophobes: Also present within the cords are small, pale-staining chromophobes with sparse cytoplasm
- **Pars Intermedia** A narrow, intermediate zone called the pars intermedia is visible, containing cellular cords and characteristic follicles filled with a pinkish, gelatinous colloid material.
- **Pars Nervosa** Components: The posterior part (pars nervosa) lacks glandular cell cords and is instead composed of unmyelinated nerve fibers
- Herring Bodies: Within the pars nervosa, faint, amorphous eosinophilic expansions known as Herring bodies can be identified



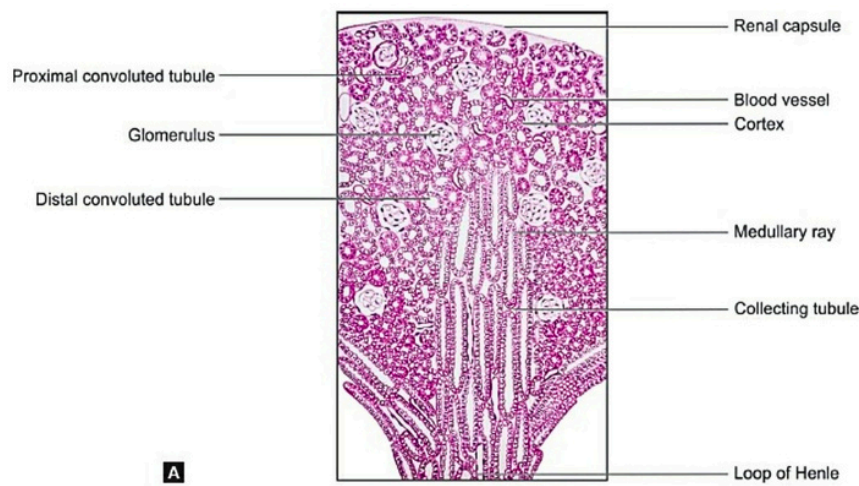
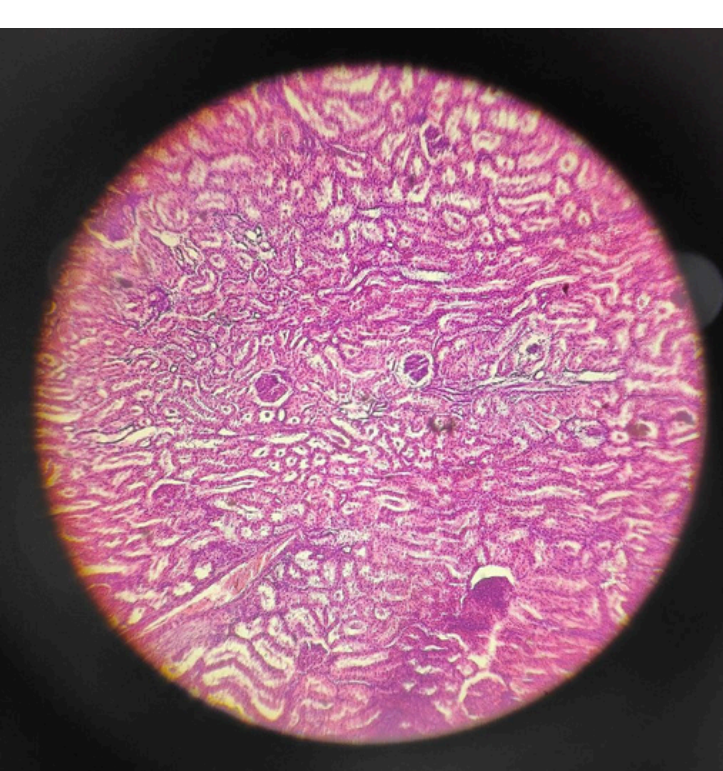


# Pancreas

- **Interlobular duct lined by simple cuboidal epithelium**
- **Islets of Langerhans:** The endocrine pancreas is composed of rounded collections of polygonal cells embedded within the exocrine part, separated by a thin layer of reticular tissue.
- **Alpha Cells:** Positioned toward the periphery, they make up 20% of islet cells, possess high electron-density granules, and secrete glucagon.
- **Beta Cells:** Form about 70% of the islet cells, tend to lie near the center, contain larger granules secrete insulin.
- **Delta Cells (D cells):** Constitute 10% of islet cells, are peripherally located, stain black and produce somatostatin and gastrin.
- **Pancreatic islets** are richly supplied by autonomic nerves; noradrenaline and acetylcholine modulate islet cell secretions.

| <i>Serous salivary gland</i>                                                | <i>Pancreas</i>                                 |
|-----------------------------------------------------------------------------|-------------------------------------------------|
| Contains only exocrine part                                                 | Contains both exocrine and endocrine parts      |
| Islets of Langerhans are absent                                             | Islets of Langerhans are present                |
| Striated ducts are present                                                  | Striated ducts are absent                       |
| Centroacinar cells are absent                                               | Centroacinar cells are present within the acini |
| Myoepithelial cells are seen between the basement membrane and acinar cells | Myoepithelial cells are absent                  |



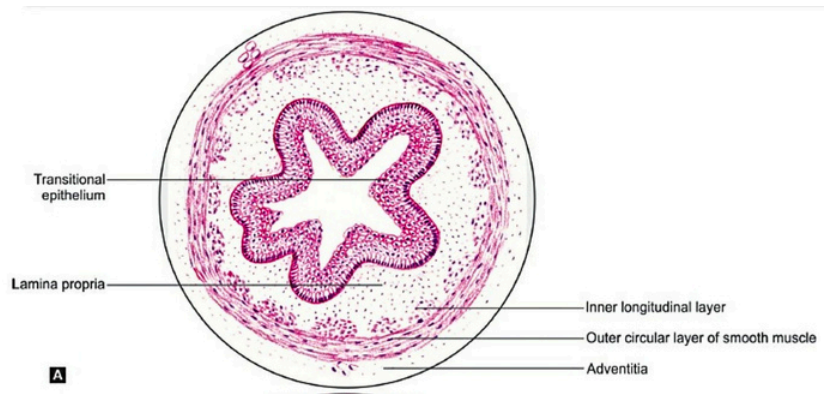


# Kidney

- **Loop of henle** is lined by **simple squamous epithelium**
- Podocytes: Specialized visceral layer cells of Bowman's capsule that possess primary processes branching into secondary processes
- **Proximal Convoluted Tubule (PCT)**: Lined by **simple cuboidal or low columnar epithelium** characterized by a prominent acidophilic cytoplasm and a dense brush border (microvilli) that reabsorbs 80% of the filtrate.
- Loop of Henle Segments: The U-shaped loop features a thin segment lined by simple squamous epithelium and a thick ascending segment lined by simple cuboidal epithelium.
- **Distal Convoluted Tubule (DCT)**: Lined by **simple cuboidal epithelium** that completely lacks a brush border, giving it a wider, cleaner-looking lumen than the PCT.
- Collecting Ducts: Lined by simple cuboidal to columnar epithelium consisting of two distinct cell types: Principal (clear) cells and Dark (intercalated) cells.

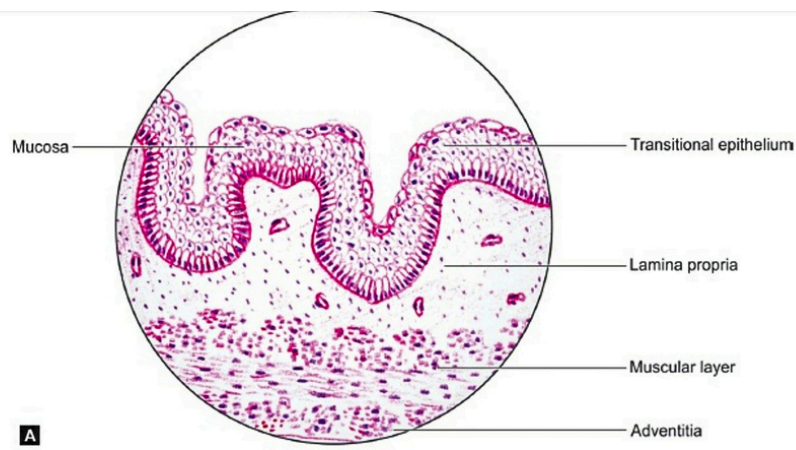
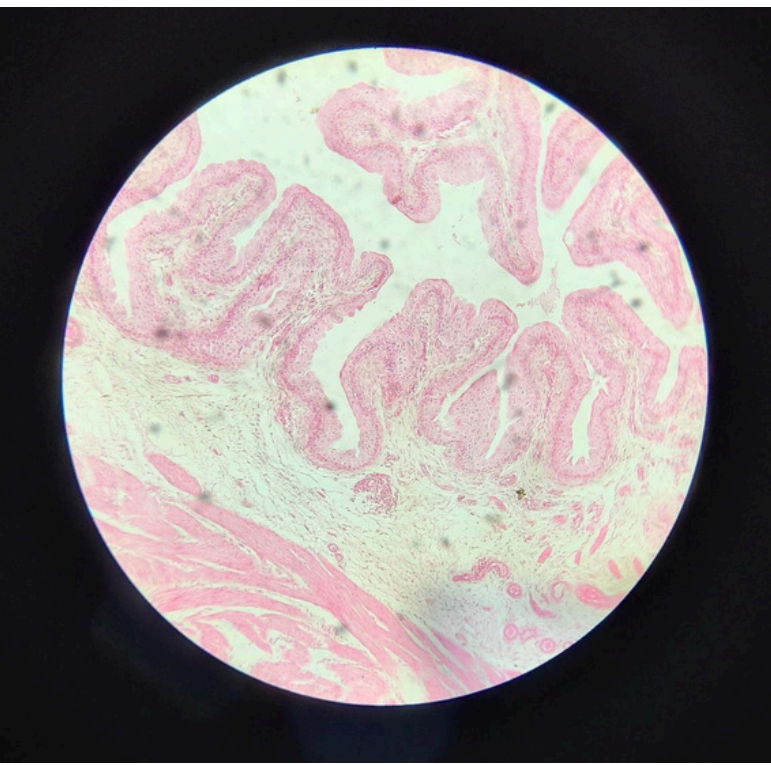
| <b>Proximal convoluted tubule (PCT)</b>                             | <b>Distal convoluted tubule (DCT)</b>                              |
|---------------------------------------------------------------------|--------------------------------------------------------------------|
| Begins from the Bowman's capsule                                    | Begins from ascending limb of loop of Henle                        |
| Lengthy so more number of PCT sections are seen in cortex of kidney | Shorter so less number of DCT are seen in cortex of kidney section |
| Continuous with loop of Henle                                       | Continuous with collecting duct                                    |
| Lined by cuboidal epithelium with microvilli                        | Lined by cuboidal epithelium without microvilli                    |
| Cytoplasm is more acidophilic                                       | Cytoplasm is less acidophilic/ basophilic                          |
| More convoluted                                                     | Less convoluted                                                    |
| Main function is reabsorption                                       | Main function is secretion                                         |





# Ureter

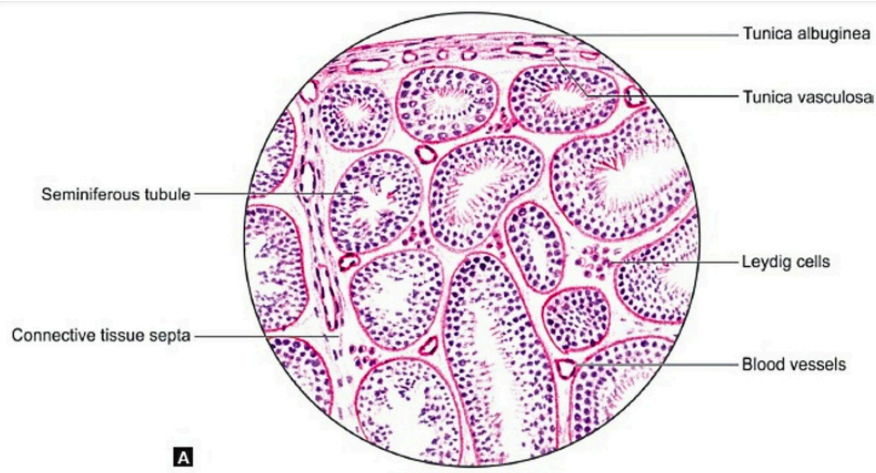
- **Star-Shaped Lumen:** The mucosa forms prominent longitudinal folds, giving the lumen a characteristic star-shaped appearance in transverse section.
- **Lining Epithelium:** It is lined by **transitional epithelium (urothelium)** that is 4 to 5 cells thick to accommodate stretching.
- **Lamina Propria:** Lies beneath the epithelium and consists of dense, irregular connective tissue, but lacks a muscularis mucosae.
- **Muscle Layer Arrangement (Upper 2/3):** Unlike the GI tract, the smooth muscle coat has a reversed arrangement: an **inner longitudinal and an outer circular layer**.
- **Lower 1/3 Muscle Layer:** In its lower one-third, a third layer (outer longitudinal) is added outside the circular coat.



# Urinary Bladder

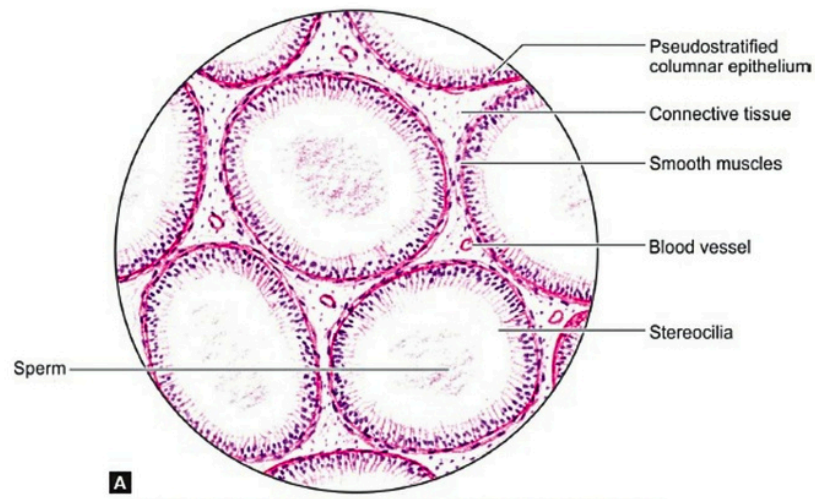
- **Urothelium & Umbrella Cells:** It is lined by urine-proof transitional epithelium featuring surface polyhedral cells called umbrella cells that spread out to allow distension.
- **No Muscularis Mucosae:** The mucous membrane of the urinary bladder completely **lacks a muscularis mucosae layer**.
- **Trigone Distinction:** The submucosal layer consists of loose areolar tissue everywhere except at the trigone, where the mucosa adheres directly to the muscle.
- **Detrusor Muscle Layers:** The thick muscular coat, or detrusor muscle, is formed by three interlaced layers: internal longitudinal, middle circular, and external longitudinal.
- **Serosa vs. Adventitia:** The superior surface is covered by peritoneum (serous layer), while the inferior part is covered by fibroelastic connective tissue (adventitia).





# Testis

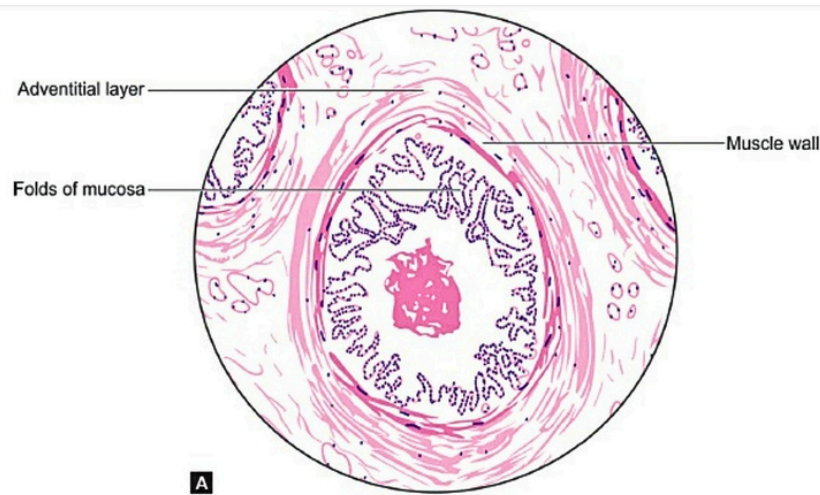
- **Capsule Layers:** The testis is surrounded from outside inwards by three distinct layers: the visceral layer of **tunica vaginalis**, **tunica albuginea**, and **tunica vasculosa**.
- **Lobular Structure:** Radiating fibrous septa divide the testis substance into 200–300 conical lobules, with each lobule housing 2–3 highly convoluted **seminiferous tubules**.
- **Sertoli Cells:** These are tall, pyramidal/columnar cells with a basally located, oval nucleus and a prominent nucleolus, whose bases rest directly on the basement membrane.
- **Leydig (Interstitial) Cells:** Large, round or polyhedral steroid-secreting cells lying in the connective tissue between seminiferous tubules, featuring an eccentric nucleus and a foamy cytoplasm.



# Epididymis

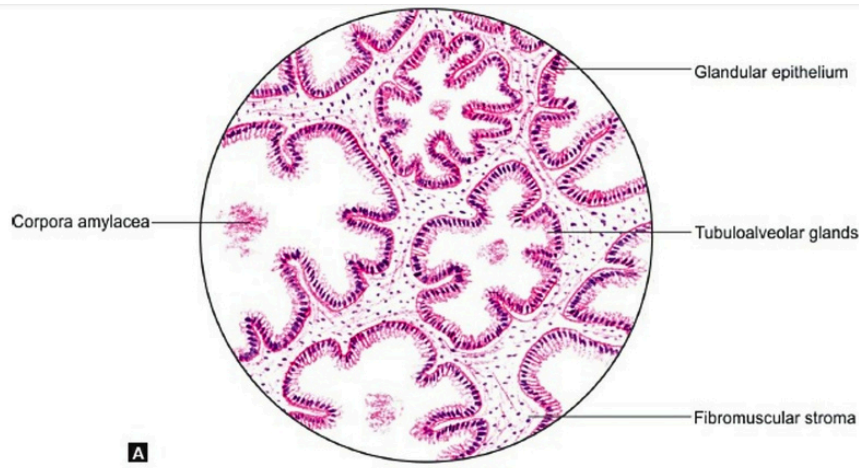
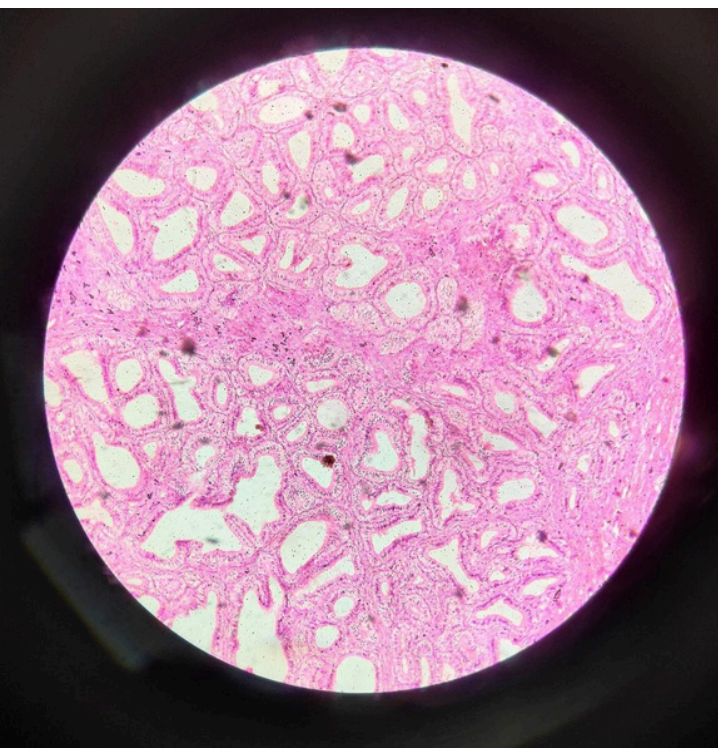
- **Lining Epithelium:** The main duct of the epididymis is lined by **pseudostratified columnar epithelium** composed of two distinct cell types: tall principal cells and short basal cells.
- **Stereocilia:** The luminal surfaces of the tall principal cells bear long, branched, irregular, and non-motile microvilli known as stereocilia.
- **Basal Cells as Stem Cells:** The short, dark-staining basal cells do not reach the lumen and function as stem cells to replace old or damaged epithelial cells.
- **Muscular Layer Shift:** The **smooth muscle layer thickens gradually from the head to the tail**, eventually organizing into distinct inner circular and outer longitudinal layers in the tail region.





# Vas Deferens

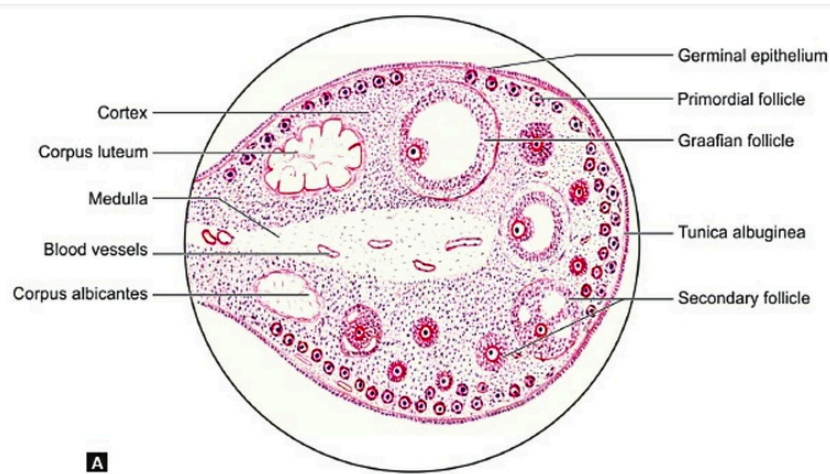
- **Epithelial lining - pseudostratified columnar**
- Lamina Propria Support: The lining epithelium rests on a cellular lamina propria that contains an abundance of elastic fibers.
- **Extremely Thick Muscle Coat:** A defining feature under the microscope is its remarkably thick muscular wall, which is highly developed relative to the narrow lumen.
- Three Muscle Layers: The smooth muscle coat is uniquely organized into three distinct layers: an inner longitudinal layer, a middle circular layer, and an outer longitudinal layer.
- Adventitia: The outermost layer is composed of **fibroelastic connective tissue** carrying prominent blood vessels and nerves..



# Prostate

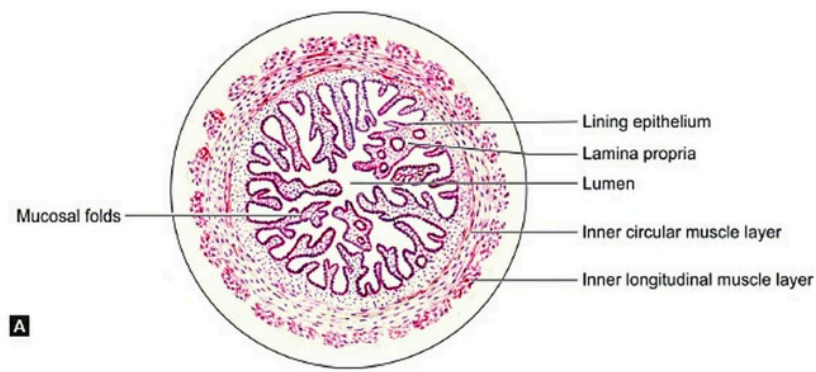
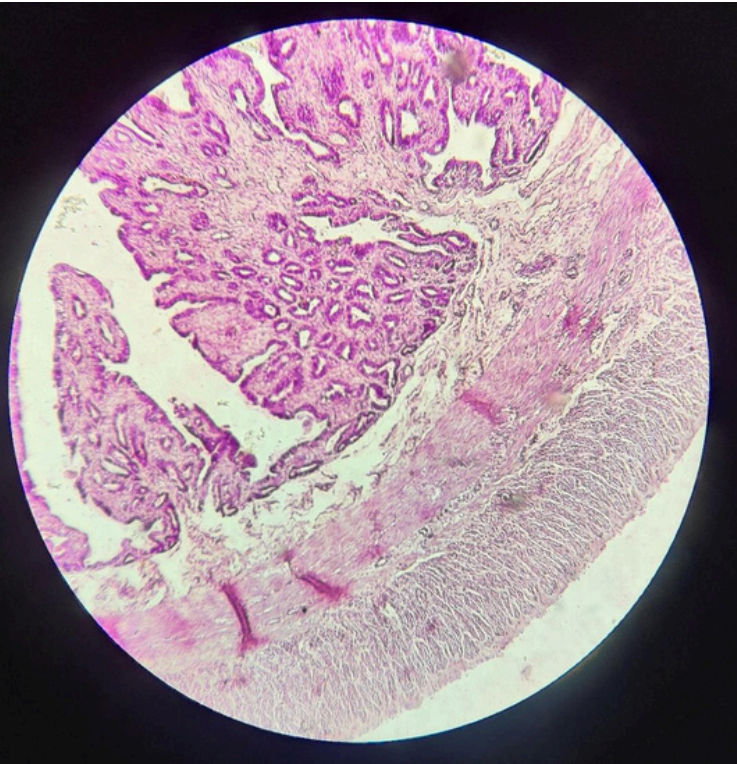
- Glandular Architecture: The glandular part is made up of 30–50 compound tubuloalveolar glands embedded directly within a **distinctive fibromuscular stroma**.
- Lining Epithelium: The secretory units or **follicles** are seen as numerous spaces lined by **simple columnar epithelium**.
- Mucosal Folding: The lining epithelium is thrown into numerous branching folds supported by underlying connective tissue.
- Concentric Layering: The glands are structurally arranged in three concentric layers around the urethra: inner mucous, intermediate submucous, and outer main prostatic glands.
- Fibromuscular Stroma: The conspicuous stroma contains a dense mixture of collagen fibers and smooth muscle, forming internal septa continuous with the outer fibrous capsule.





# Ovary

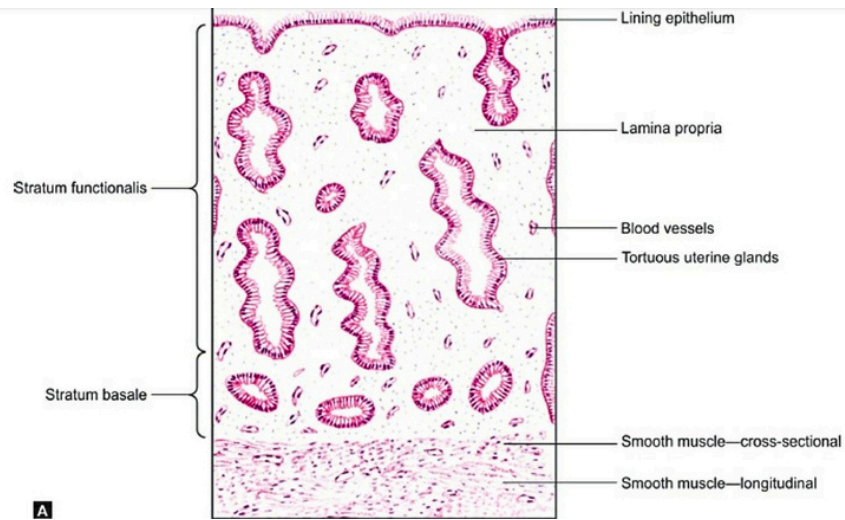
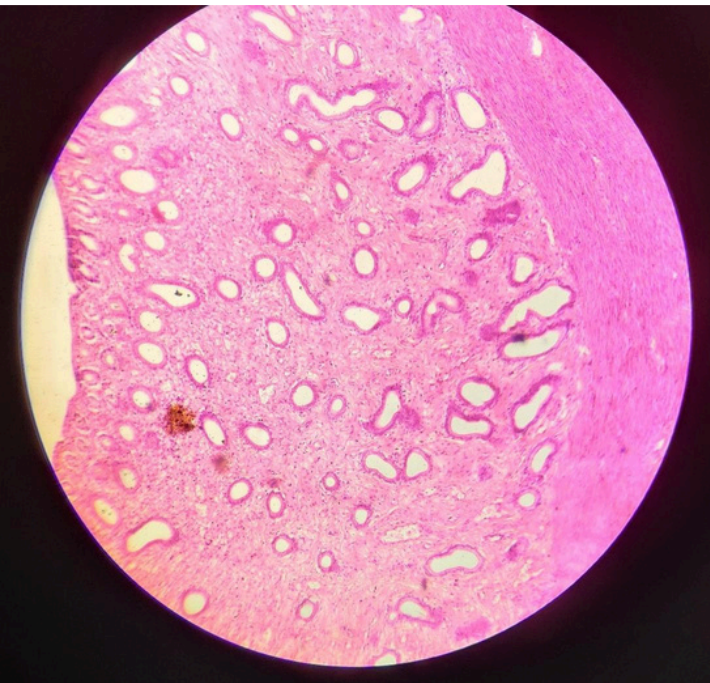
- **Tunica Albuginea:** Beneath the epithelium lies a dense, irregular collagenous connective tissue capsule that is thinner and less dense than that of the testis.
- **Zonal Division:** The inner substance of the gland is distinctly divided into an **outer cortex containing follicles** and an **inner vascular medulla of loose connective tissue**.
- **Primordial Follicles:** These are the smallest and simplest follicles located at the periphery of the cortex, consisting of a primary oocyte surrounded by a single layer of flattened follicular cells.
- **Zona Pellucida:** Appears as a prominent, homogeneous membrane between the growing oocyte and the surrounding cuboidal follicular cells in the primary follicle stage.
- **Graafian Follicle Architecture:** The mature stage features an eccentric primary oocyte projected into a fluid-filled cavity (antrum) by a specialized group of granulosa cells called the cumulus oophorus.
- **Corpus Luteum Transition:** Post-ovulation, the ruptured follicle initially fills with blood to form the corpus hemorrhagicum before cells expand and fill with a yellow pigment called lutein.



## Fallopian Tube

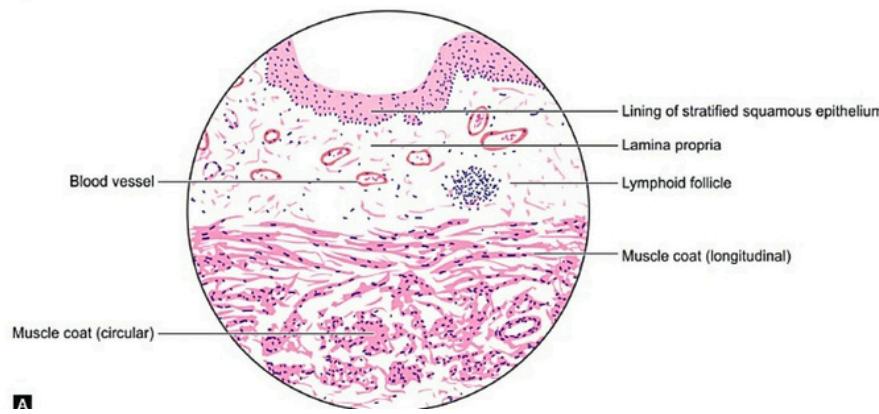
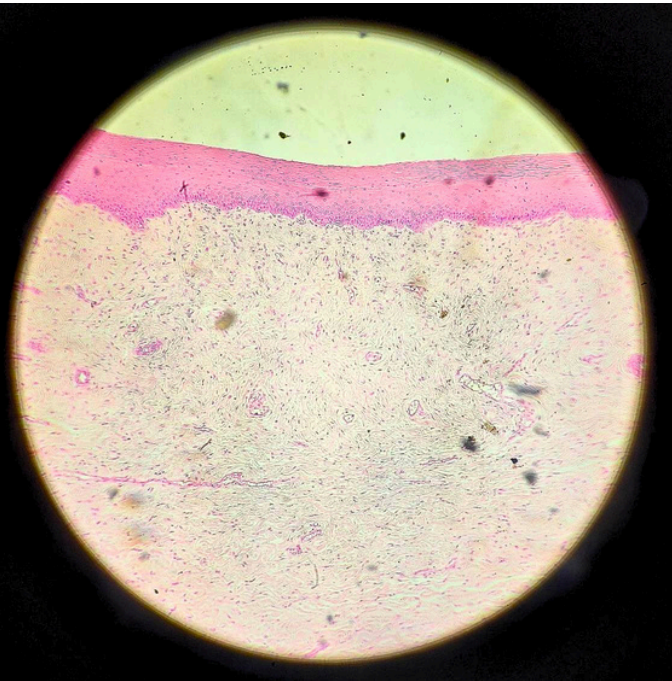
- **Branching Mucosal Folds:** The mucosa shows three to six longitudinal folds that branching into secondary and tertiary folds, which almost completely fill the lumen and are most conspicuous in the ampulla.
- **Lining Epithelium - simple columnar epithelium resting.**
- **Ciliated Cells:** These cells possess cilia that physically assist in transporting and moving the ova along the tube.
- **Peg Cells (Secretory Cells):** Non-ciliated cells with microvilli that contain secretory granules to provide vital nutrition to the fertilized ovum and spermatozoa.
- **Muscular Layer Arrangement:** Formed of smooth muscle arranged into an inner circular layer and an outer longitudinal layer, with circular muscle being abundant at the isthmus to act as an anatomical sphincter.





# Uterus

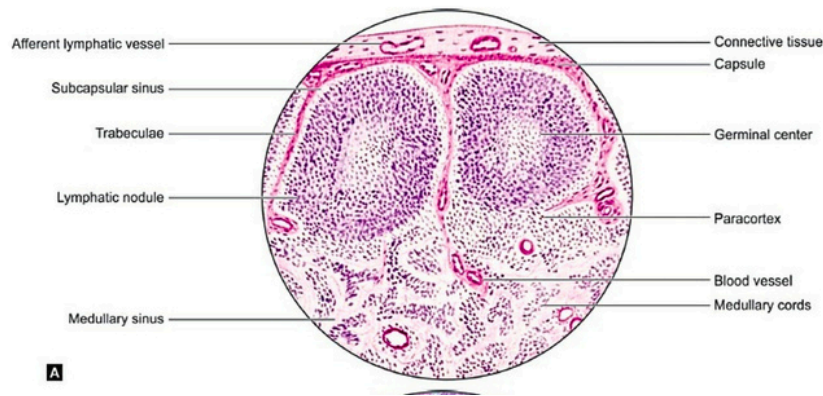
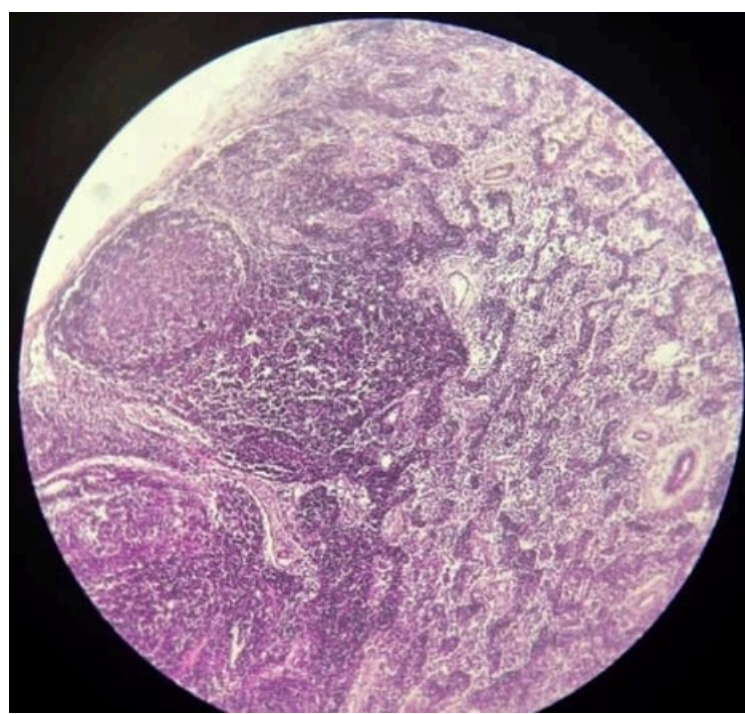
- Wall Layers: From within outwards, the very thick wall of the uterus consists of three layers: the **endometrium, myometrium, and perimetrium**.
- **Lining Epithelium is simple columnar**
- **Lamina Propria/Stroma:** The endometrial stroma contains specialized embryonic stromal cells, numerous blood vessels, lymphatics, nerves, and tubular uterine glands.
- Uterine Glands: The mucosa contains numerous tubular glands lined by simple columnar epithelium that extend all the way up to the inner surface of the myometrium.
- Menstrual Cyclic Changes: The functional layer undergoes cyclical changes and is cast off each month during menstruation, whereas the basal layer remains practically unaltered to regenerate the tissue.



# Vagina

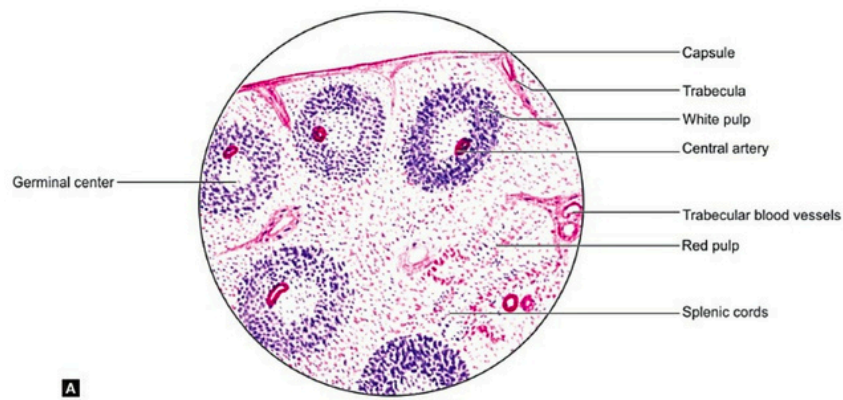
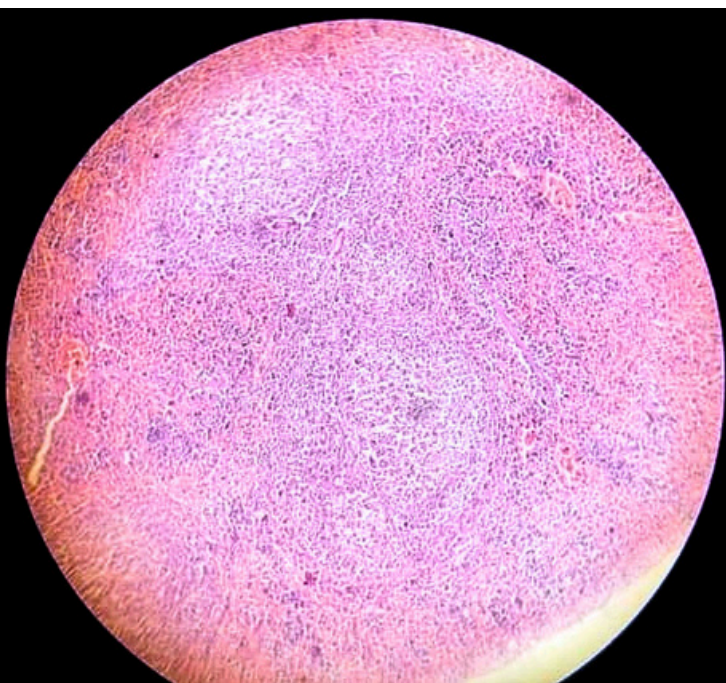
- **Lining Epithelium: stratified squamous nonkeratinized epithelium.**
- Lamina Propria: Composed of loose connective tissue containing a plexus of blood vessels, lymphatics, nerves, and a rich network of elastic fibers.
- Smooth Muscle Arrangement: The **muscle coat consists of a thicker outer longitudinal layer and a much thinner inner circular layer.**
- External Sphincter: The lower end of the vagina is surrounded by striated muscle fibers (bulbospongiosus muscle) forming an external sphincter.
- Adventitia vs. Serosa: Surrounded primarily by a fibrous adventitia, a serous layer (derived from peritoneum) covers only the posterior fornix.





# Lymph Node

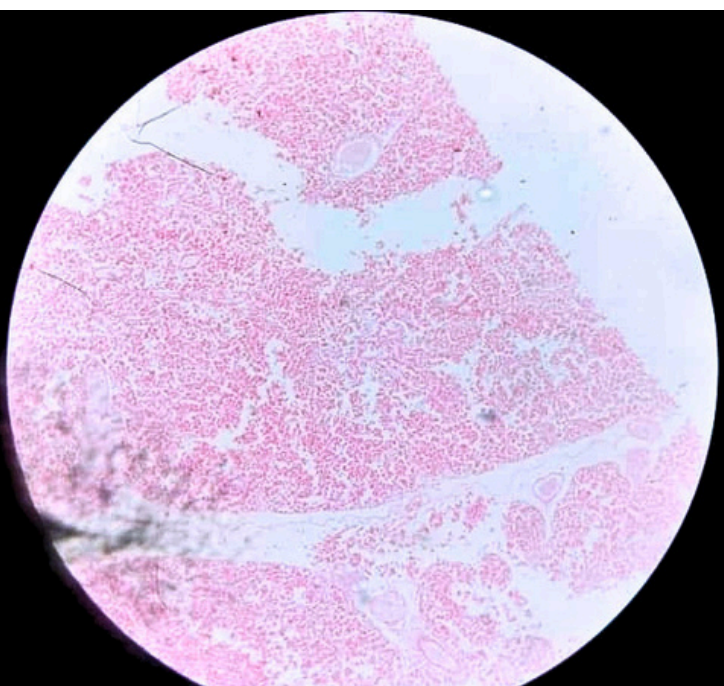
- **Capsule and Subcapsular Sinus:** The node is invested by a fibrous capsule separated from the cortex by a subcapsular space/sinus that receives afferent lymph vessels.
- **Lymphatic Follicles (Nodules):** The superficial cortex features primary nodules (aggregations of small lymphocytes) and secondary nodules (containing a central germinal center indicating antigen response).
- **Medullary Cords:** Branching and anastomosing cords of lymphoid tissue within the medulla that house B-lymphocytes, T-lymphocytes, macrophages, dendritic cells, and plasma cells.
- **Medullary Sinuses:** Dilated spaces lined by a discontinuous endothelium that separate the medullary cords, act as a lymph filter, and converge at the hilum as the efferent vessel.



# Spleen

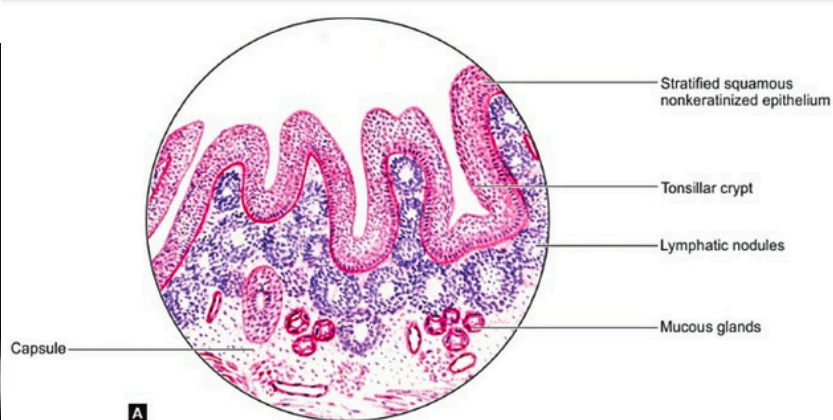
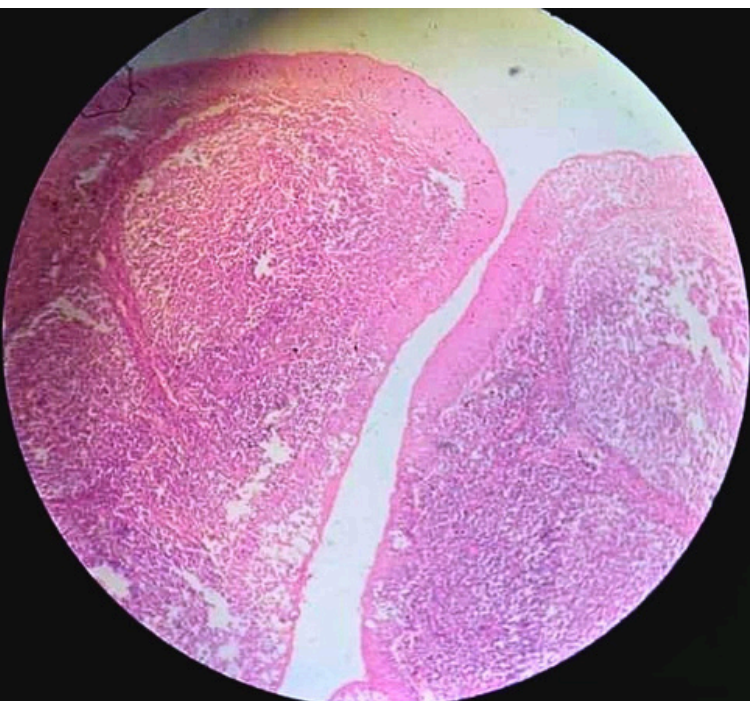
- **White Pulp** : Appears highly basophilic on H&E stains because it consists of dense aggregations of lymphocytes forming the periarterial lymphatic sheath (PALS) around a central arteriole.
- Malpighian Bodies: Localized expansions of the PALS containing large B-lymphocyte germinal centers formed in response to antigens are called splenic nodules or Malpighian bodies.
- **Red Pulp** Composition: Comprises the remaining tissue matrix and is made up of splenic cords (of Billroth) containing meshworks of reticular cells, macrophages, and blood cells.
- Open vs. Closed Circulation: Penicillar arterioles either empty directly into the meshwork of the red pulp (open circulation, seen in humans) or connect directly to the sinusoids (closed circulation).





# Thymus

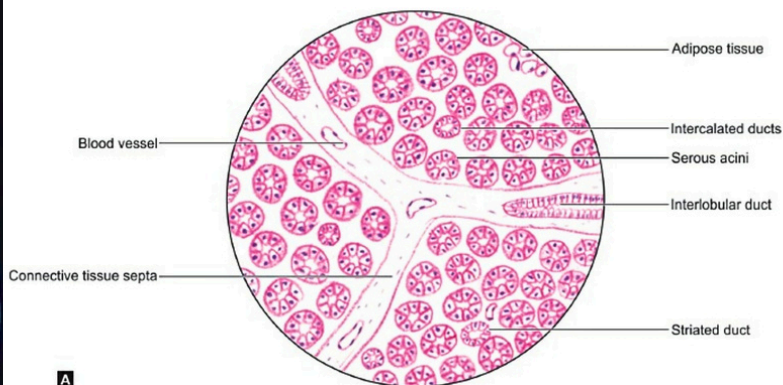
- **Hassall's Corpuscles:** Located in the medulla, these diagnostic, pink-stained, rounded structures.
- **Lobular Architecture:** Connective tissue septa project inward from the capsule to divide the organ into incompletely separated lobules.
- **Cortex vs. Medulla:** Each lobule has a **darkly stained outer cortex** packed with developing T-lymphocytes (thymocytes) and an inner pale medulla that is continuous between adjoining lobules.
- **Thymic Nurse Cells:** Type 2 and Type 3 epitheliocytes compartmentalize the cortex and act as nurse cells that destroy self-reactive, autoimmune T-lymphocytes.



# Palatine Tonsil

- **Epithelial Covering:** The free surface is covered by **stratified squamous non-keratinized epithelium**
- **Tonsillar Crypts :** The surface epithelium invaginates deeply into the underlying lymphoid tissue to form 10 to 15 elongated, blind-ending tonsillar crypts.
- **Lymphatic Nodules:** The underlying lamina propria contains numerous dense, round aggregations of lymphocytes called lymphatic nodules (follicles).
- **Germinal Centers:** Most lymphatic nodules exhibit a lighter-staining central area known as a germinal center, indicating active B-lymphocyte proliferation.
- **Absence of Sinuses:** Unlike a lymph node, the palatine tonsil completely lacks subcapsular or medullary sinuses and does not possess afferent lymphatic vessels.

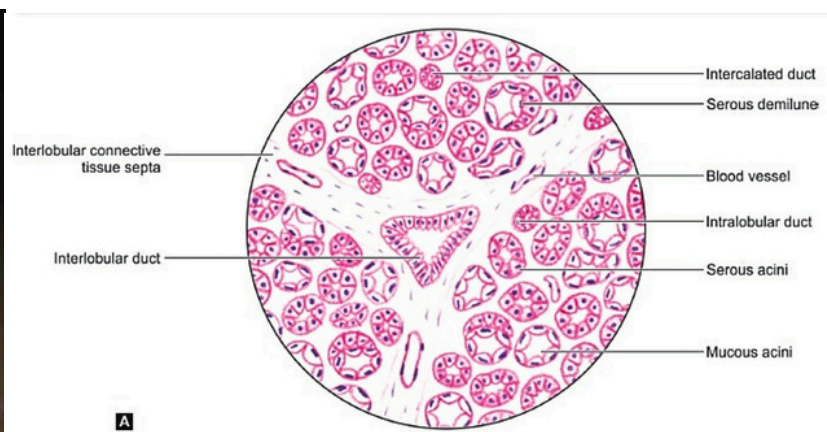
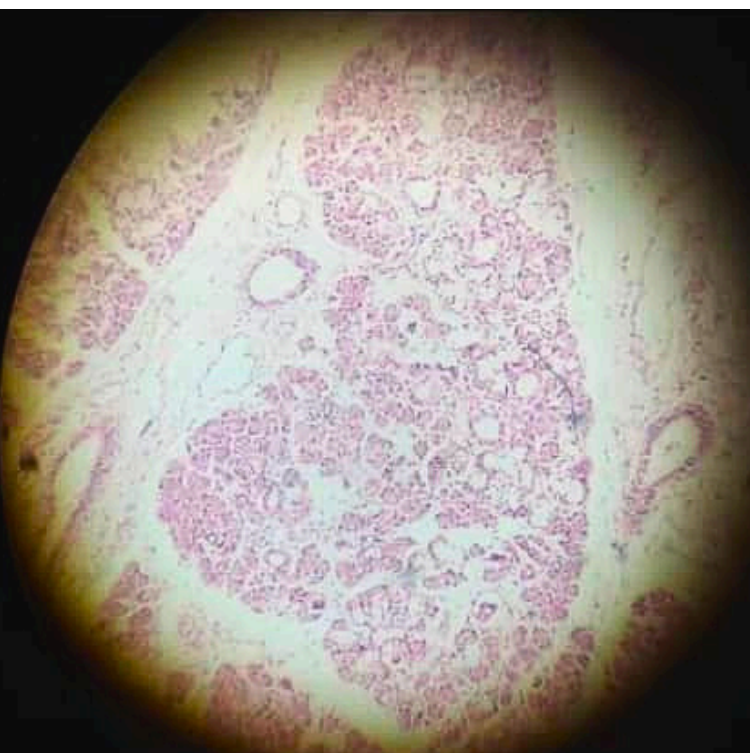




A

# Parotid Gland

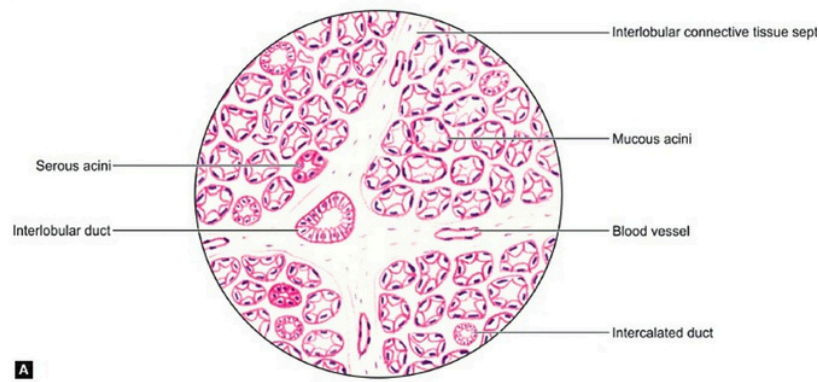
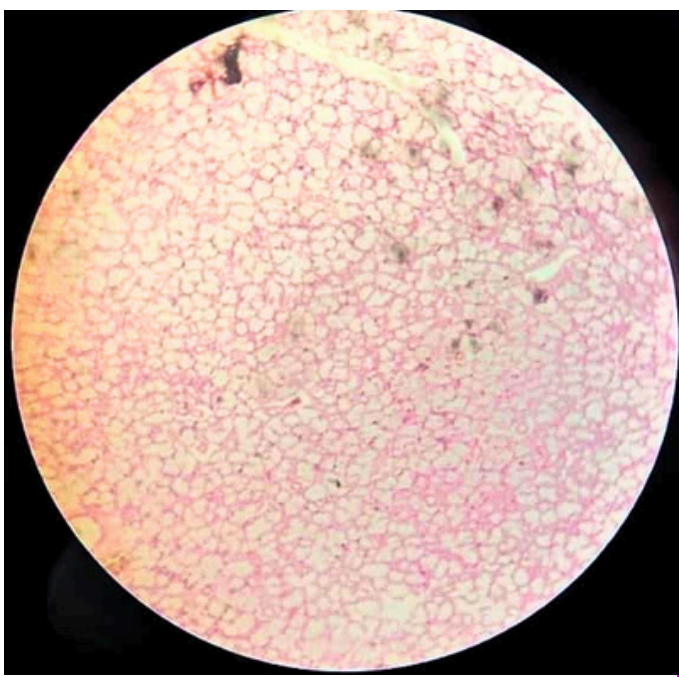
- **Purely serous salivary gland** structurally classified as a compound tubuloacinar (or racemose) gland.
- **Stensen's Duct Location**: Its main excretory duct is Stensen's duct, which opens into the vestibule of the mouth opposite the second upper molar tooth.
- **Serous Acini** Appearance: Lobules feature pyramid-shaped secretory cells with spherical nuclei at the base of their slightly basophilic cytoplasm.
- **Myoepithelial Cells**: The acini are surrounded by thin, contractile myoepithelial cells located between the basement membrane and acinar cells to help express secretions.
- **Progressive Epithelial Changes**: The duct lumen gets wider and its epithelium taller, changing from columnar to pseudostratified or stratified columnar in large lobar ducts near their termination.



## Submandibular Gland

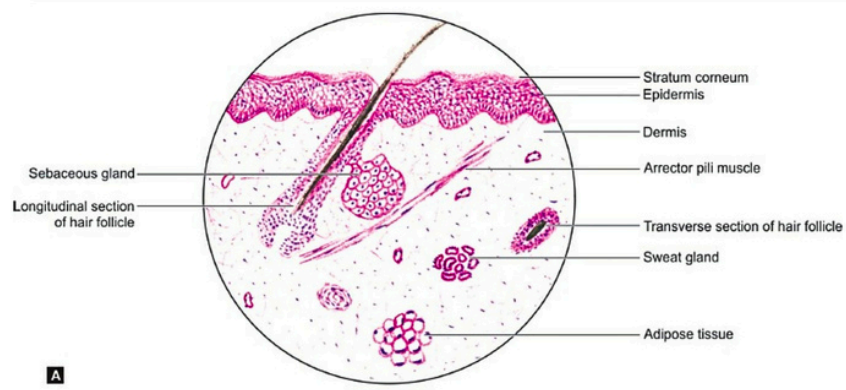
- **Mixed salivary gland** containing both serous and mucous acini, but **serous acini predominate**.
- **Serous Acini Features:** Composed of smaller, darker-stained pyramidal cells with spherical basal nuclei and apical secretory granules.
- **Mucous Acini Features:** These acini are larger with a wide lumen, featuring columnar cells with pale/colorless cytoplasm and flattened nuclei pressed against the base.
- **Serous Demilunes:** A hallmark diagnostic feature where mucous acini are capped by crescent-shaped clusters of serous cells called demilunes.
- **Demilune Function:** These serous demilune caps are thought to empty their secretory products into the highly convoluted intercellular spaces between adjacent mucous cells..





# Sublingual Gland

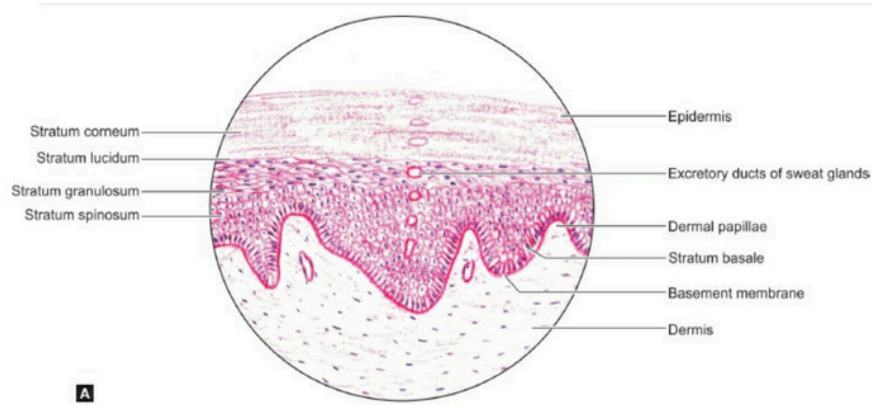
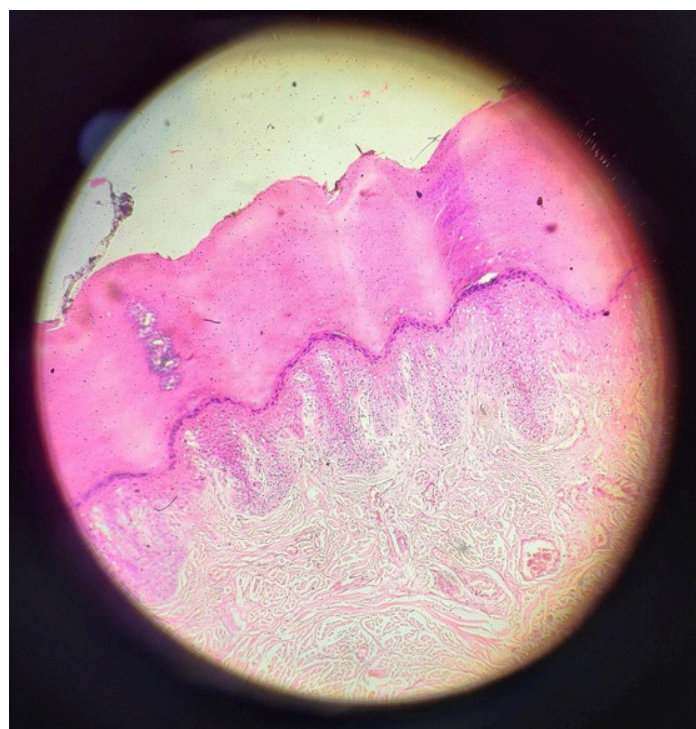
- **Mixed Gland** Nature: It is structurally classified as a compound, mixed tubuloacinar gland.
- **Mucous Predominance**: Unlike the parotid and submandibular glands, mucous acini are predominantly seen, appearing as conspicuous, light-stained structures.
- **Poorly Visible Intercalated Ducts**: The intercalated ducts are short or entirely absent, making them difficult to visualize in a typical histological section.
- **Nonstriated Intralobular Ducts**: In contrast to other major salivary glands, nonstriated intralobular excretory ducts are more prevalent here.
- **Abundant Stroma**: The interlobular connective tissue septa are highly abundant in the sublingual gland compared to the other two major salivary glands.



# Thin Skin

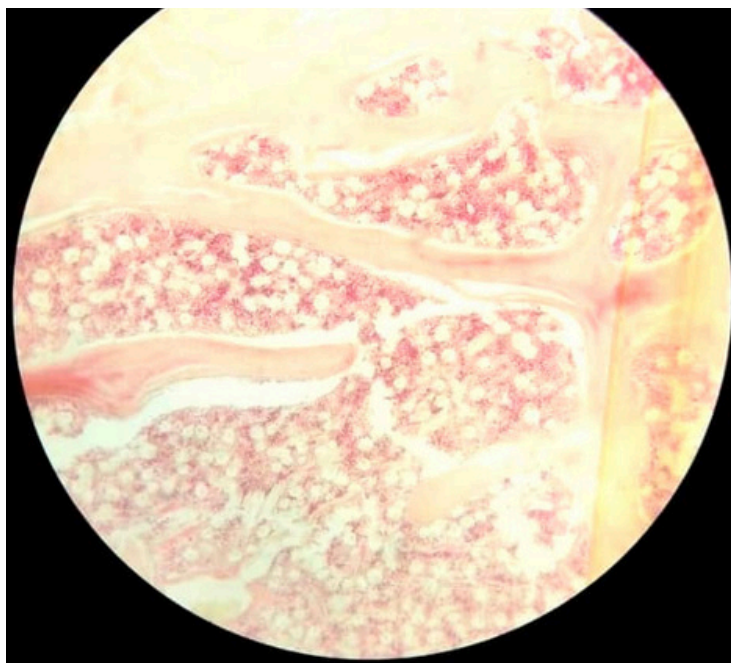
- **Lined by keratinized stratified squamous epithelium**
- **Narrow Stratum Corneum:** The outermost layer (stratum corneum) is very thin, loose, and often appears flaky or wispy on the slide.
- **Presence of Hair Follicles:** The dermis is heavily characterized by the presence of numerous hair follicles cut in various planes of section.
- **Sebaceous Glands:** Features prominent, pale-staining, multi-lobed sebaceous glands (holocrine glands) closely associated with and opening into the hair follicles.
- **Arrector Pili Muscles:** Bundles of smooth muscle fibers called arrector pili muscles are clearly visible, extending obliquely from the connective tissue sheath of hair follicles to the papillary dermis.





# Thick Skin

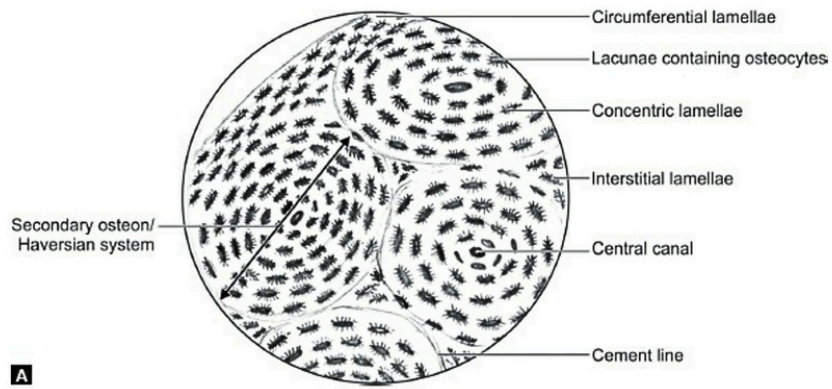
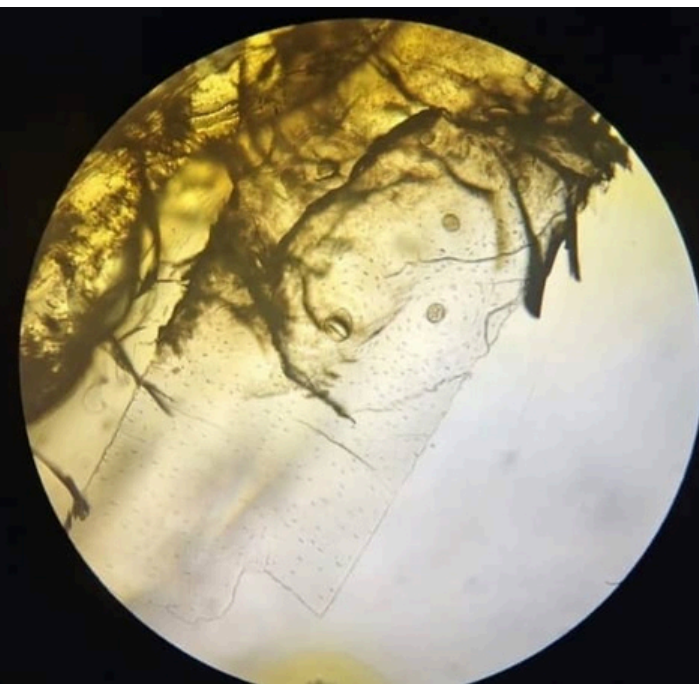
- **Lined by keratinized stratified squamous epithelium.**
- **Stratum Corneum:** The topmost layer (stratum corneum) is exceptionally thick, consisting of multiple layers of dead, flattened, fully keratinized enucleated cells.
- **Presence of Stratum Lucidum:** Contains a distinct, clear, translucent layer called the stratum lucidum located just beneath the corneum (absent or poorly visible in thin skin).
- **Prominent Stratum Granulosum:** Displays a highly prominent stratum granulosum containing 3–5 layers of flattened cells filled with dark, basophilic keratohyalin granules.
- **Abundant Sweat Glands:** The underlying dermis is packed with coiled secretory units and ducts of merocrine (eccrine) sweat glands.



## Spongy Bone

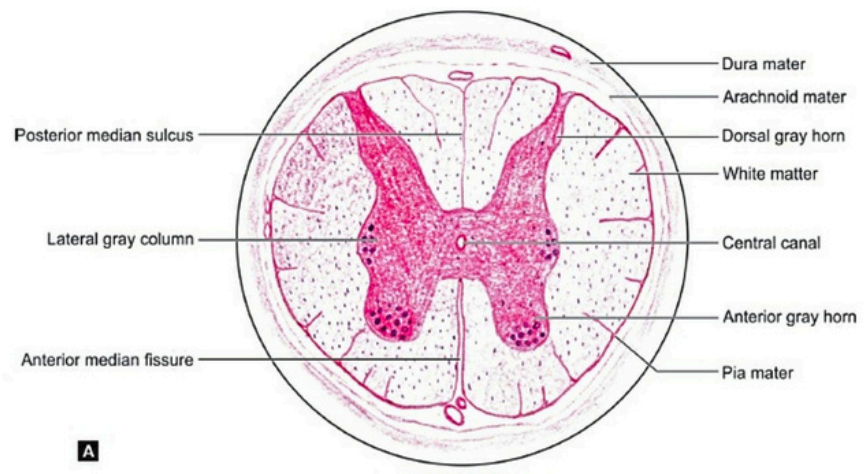
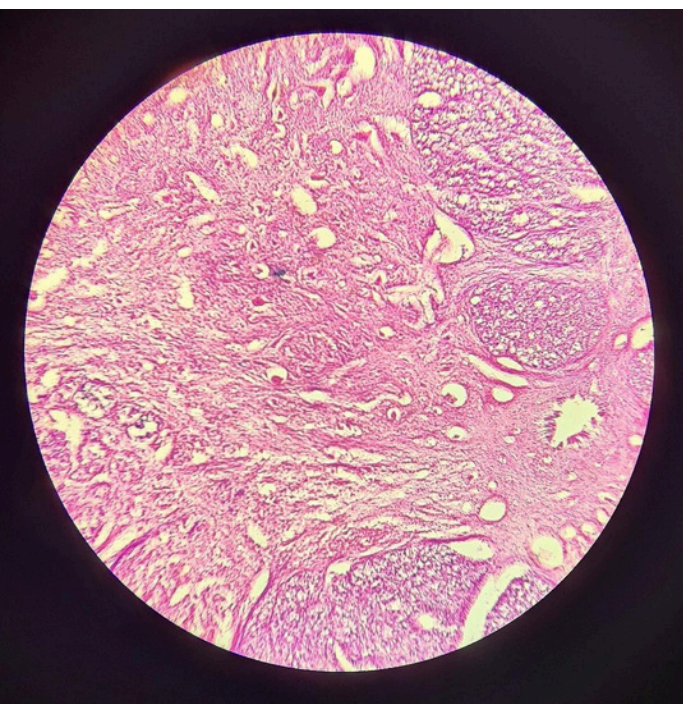
- **No Haversian System:** A critical diagnostic distinction from compact bone is that spongy bone completely lacks a Haversian system.
- **Trabecular Framework:** It is characterized by a sponge-like, trabeculated network made up of tiny bony rods or plates called **trabeculae**.
- **Marrow Spaces:** The branching trabeculae enclose wide, numerous structural spaces that are filled with bone marrow.
- **Nutrition Source:** Because it lacks central canals, the osteocytes receive their nutrition via radiating canaliculi directly from the blood vessels present in the surrounding bone marrow.
- **Endosteal Lining:** The trabeculae are covered externally by a vascular layer of endosteum containing osteoblasts, osteoclasts, and osteoprogenitor cells.





# Compact Bone

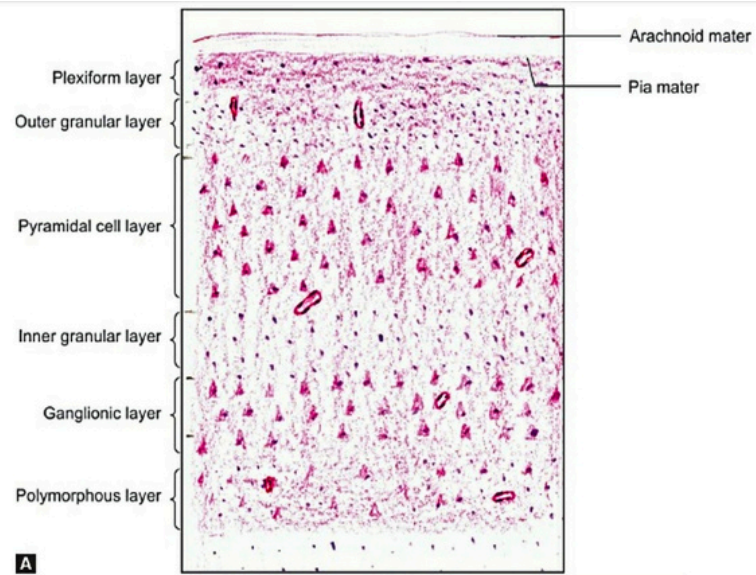
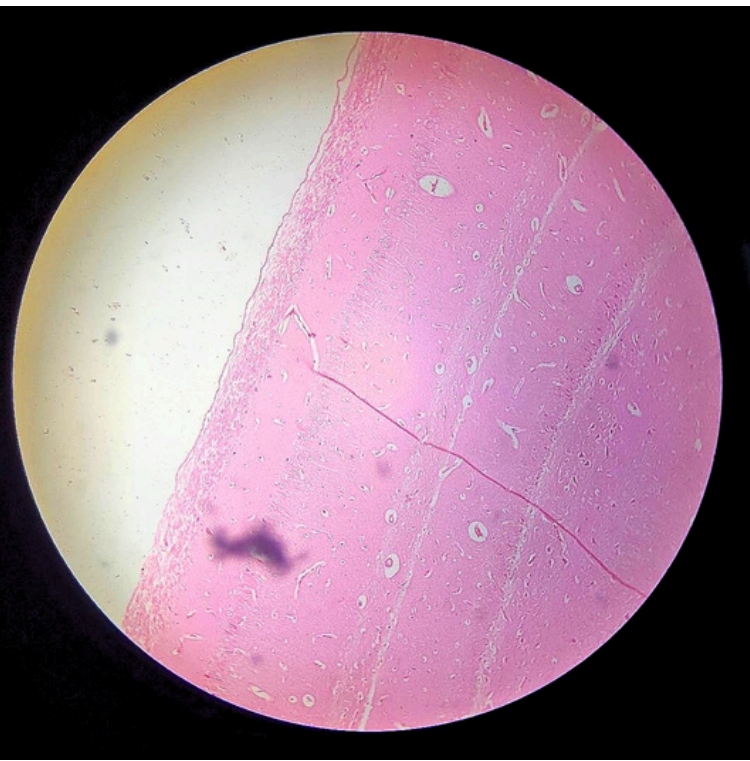
- **Concentric Lamellae:** These are rings of mineralized bone matrix arranged in concentric layers around the narrow central **Haversian canal**.
- **Interstitial Lamellae:** Found in the angular spaces between intact osteons, representing the remnants of older, partially destroyed osteons.
- **Lacunae and Osteocytes:** Small, flattened spaces scattered between adjacent lamellae, with each lacuna harboring a single osteocyte.
- **Canaliculi Network:** Fine micro-channels radiating outward from each lacuna containing cytoplasmic processes of osteocytes that interconnect to facilitate metabolic diffusion.



# Spinal Cord

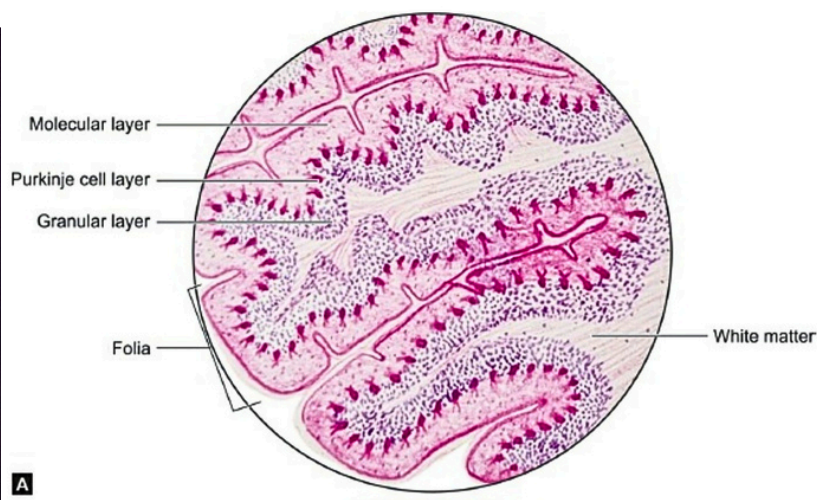
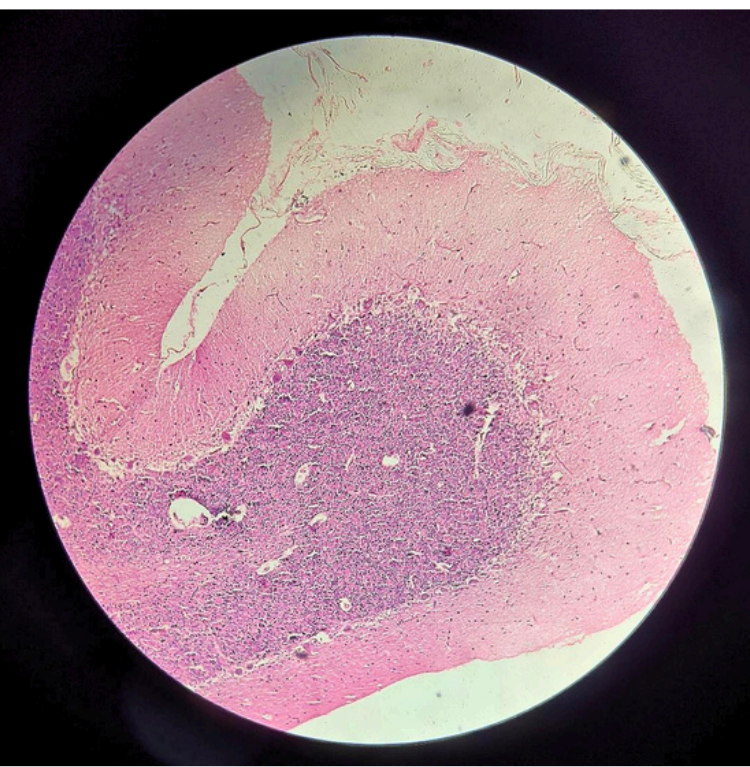
- **Zonal Arrangement:** On cross-section, the spinal cord uniquely presents an **outer white matter** zone and a **central, H-shaped inner gray matter**
- **Gray Matter Horns:** The gray matter features a pair of broad, short anterior/ventral horns and narrow, elongated posterior/dorsal horns.
- **Lateral Gray Column:** In the thoracic and upper lumbar regions, a small lateral gray column projects outwards from the base of the anterior horn.
- **Central Canal & Ependyma:** The gray commissure connecting the two halves is traversed by the central canal, which contains CSF and is lined by ependyma.
- **White Matter Funiculi:** The nerve fibers of the white matter are structurally organized into three columns on each side: the anterior, lateral, and posterior funiculi.





# Cerebrum

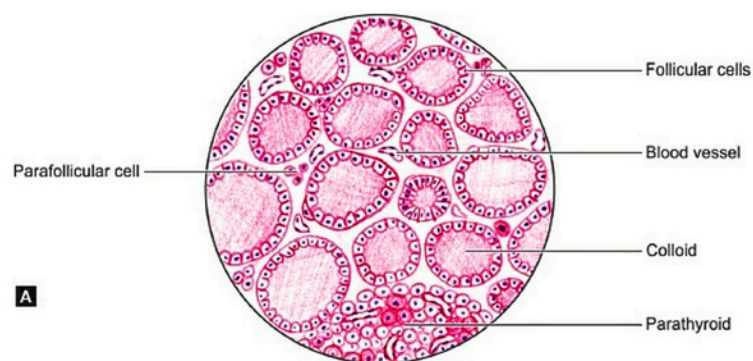
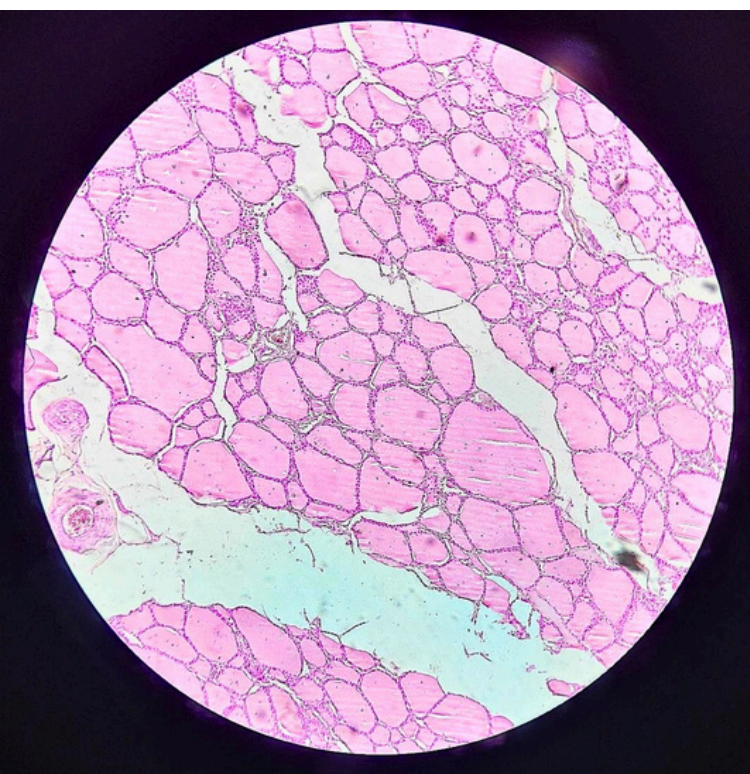
- **Gray matter** containing neuronal cell bodies forms the outer surface (cortex) covering the inner core of **white matter**.
- **Pyramidal Cells:** These cell bodies are triangular/pyramidal in shape, constitute two-thirds of all cortical neurons, and have apices generally directed toward the cortical surface.
- **Stellate (Granule) Neurons:** Relatively small, multipolar Type II neurons with short axons that end within the cortex, making up about one-third of the total neuronal population.
- **Specialized Interneurons:** Includes Cells of Martinotti (small, multipolar neurons mostly in the sixth layer with axons running vertically upward) and Horizontal Cells of Cajal (found in the first layer with neurites extending tangentially).



## Cerebellar Cortex

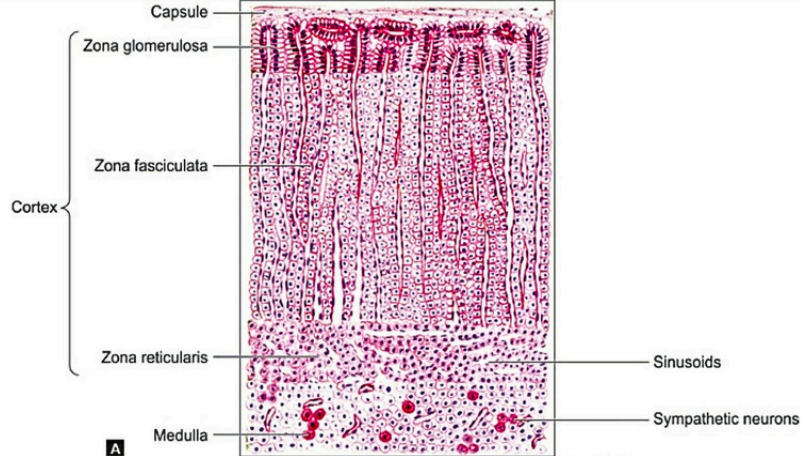
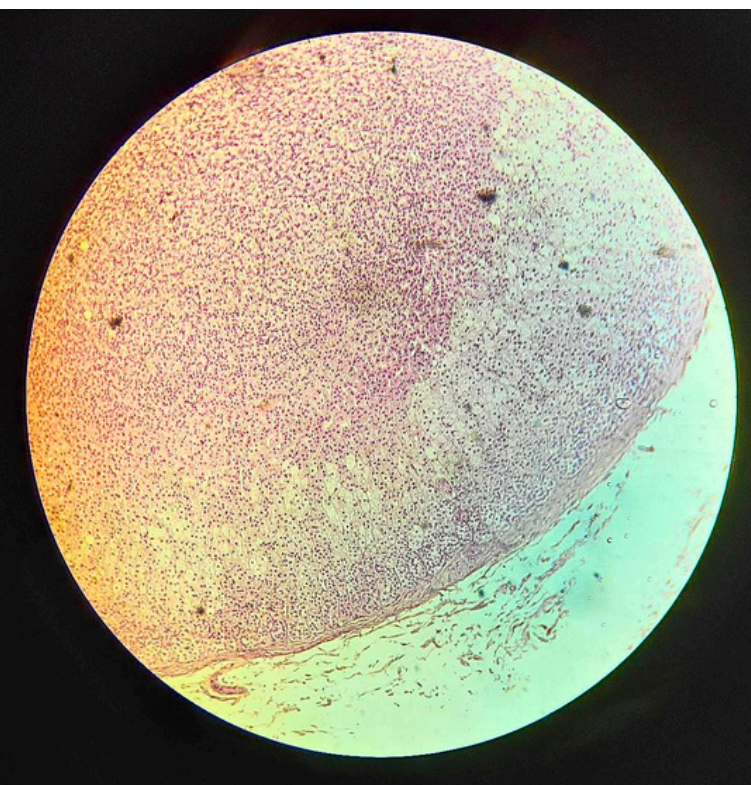
- **Surface Folia:** The grey matter surface is thrown into numerous narrow, transverse folds called folia that are separated by deep fissures.
- **Three Distinct Layers:** From outside inward, the cortex contains an **outer molecular layer, a middle Purkinje cell layer, and an inner granular layer** resting on white matter.
- **Purkinje Cell Layer:** Features a single stratum of large, unique flask-shaped cell bodies whose axons act as the sole output neurons of the cerebellar cortex.
- **Granular Layer Architecture:** Densely packed with very small granule cells (the only excitatory neurons here) and larger Golgi cells.
- **Cerebellar Glomerulus:** A complex synaptic structure in the granular layer composed of a mossy fiber rosette, claw-like granule cell dendrites, and a Golgi cell axon terminal.





# Thyroid Gland

- **Thyroid Follicles:** The structural parenchyma of each lobule is composed of an aggregation of rounded spaces called follicles.
- **Follicles lined by simple cuboidal**
- **Follicular Cell** Function: Influenced by pituitary TSH, follicular cells synthesize and secrete triiodothyronine (T3) and tetraiodothyronine (T4/thyroxine).
- **Parafollicular Cells** (C-cells): Also called clear or light cells, these polyhedral cells with eccentric nuclei lie in groups between follicles or follicle linings, but do not reach the lumen.
- **C-cell** Function: Derived from the neural crest, parafollicular cells secrete the hormone calcitonin to lower blood calcium levels when they are high.



# Adrenal Gland

- **Zona Glomerulosa (Outermost):** Constitutes the outer one-fifth of the cortex; its small, columnar cells are arranged in inverted U-shaped cords or rounded acinus-like groups to secrete mineralocorticoids (aldosterone).
- **Zona Fasciculata (Middle):** Forms three-fifths of the cortex; its cells are arranged in straight, parallel columns separated by sinusoids.
- **Zona Reticularis (Innermost):** Forms the inner one-fifth of the cortex; its cells are arranged as a branching and anastomosing network of cords that secrete sex hormones and glucocorticoids.
- **Chromaffin Cells:** Found in the medulla, these modified postganglionic sympathetic neurons stain brown with chromium salts due to the oxidation of catecholamines.



**“Every day you choose discipline,  
you choose a better version of  
yourself.”**



**MISSION**  
— DISTINCTION —  
■ ■ ■